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On Hearing the Shape of a Two-Dimensional Analytic Drum: A Case Study of the Neumann Oval

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Abstract

The question of whether the Dirichlet spectrum determines the shape of a drum is a central problem in inverse spectral geometry. For a bounded simply connected planar domain $\Omega \subset \mathbb{R}^2$ with real-analytic Jordan boundary, we construct an entire family of boundary operators $W(\lambda)$, called the Wronskian operators. Its kernel is linearly isomorphic to the corresponding Dirichlet eigenspace, and its regularized determinant vanishes at the Dirichlet eigenvalues. In conformal Fourier coordinates, we derive a representation of $W(\lambda)$ from the coefficients of the normalized Riemann map. We illustrate the finite construction through a forward spectral calculation and a restricted three-parameter fitting experiment for the Neumann oval with $R = 1.8$.

Keywords: Inverse spectral geometry; Dirichlet eigenvalues; analytic planar domains; Wronskian operator; Riemann mapping coefficients; Neumann oval.

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1 Introduction

Fix a bounded planar domain $\Omega \subset \mathbb{R}^2$. Its Dirichlet eigenpairs satisfy

$$\begin{cases} -\Delta u = \lambda u, & x \in \Omega, \\ u = 0, & x \in \partial\Omega, \end{cases} \qquad u \not\equiv 0. \tag{1.1}$$

The eigenvalues form a discrete sequence

$$0 < \lambda_1 \le \lambda_2 \le \cdots \longrightarrow \infty,$$

where multiplicities are included. Such eigenvalue problems arise in acoustics, structural vibration, quantum mechanics, diffusion theory, and electromagnetic waveguides. For a membrane with a fixed boundary, $\sqrt{\lambda_j}$ is proportional to a resonant frequency, while the corresponding eigenfunction describes its vibration mode. Related Laplace spectra also occur in quantum billiards and bounded resonators. Reviews of Laplacian eigenvalues and of the mathematical and physical aspects of isospectrality are given in [1, 2].

The relation between these frequencies and the geometry of Ω is a classical question in spectral geometry. In 1911, Weyl proved that their leading asymptotic distribution determines the area of a planar domain [3]. Heat-kernel expansions later showed that the spectrum also determines the boundary length and several integral curvature invariants [4]. In 1966, Kac asked whether the complete Dirichlet spectrum determines the entire shape of a planar domain up to rigid motion [5]. This question distinguishes the recovery of individual geometric invariants from the global uniqueness of the underlying domain.

Before Kac's formulation, a related obstruction was already known in the broader Riemannian setting. In 1964, Milnor used two sixteen-dimensional lattices constructed by Witt to obtain isospectral nonisometric flat tori [6]. Milnor's construction reduced isospectrality to equality of the theta series of the corresponding dual lattices and introduced an arithmetic mechanism for producing identical spectra from different geometries. Sunada developed a group-theoretic construction using finite coverings and almost-conjugate subgroups in 1985 [7]. Gordon, Webb, and Wolpert brought this mechanism to planar domains in 1992, using transplantation to produce noncongruent polygons with identical Dirichlet spectra [8]. Their example showed that spectral equality alone does not imply congruence for arbitrary planar domains.

Subsequent research focused increasingly on smooth or analytic boundaries and on geometrically restricted classes possessing strong symmetry. In the smooth planar setting, fixing the spectrum yields a C^∞-compact collection of domains [9]. For real-analytic domains, Zelditch placed the Balian–Bloch trace formula on a rigorous analytic foundation and used wave invariants associated with periodic billiard trajectories to obtain spectral determination within several symmetric classes [10, 11, 12]. Hezari and Zelditch extended this approach to higher-dimensional analytic domains with coordinate-reflection symmetries [13]. The restriction to real-analytic boundaries introduces additional complex-analytic structure. Such a boundary admits a local Schwarz function S, characterized by $S(z) = \overline{z}$ on $\partial\Omega$, and both S and the Riemann map extend across the boundary. These continuation properties are used in analytic reflection, potential theory, and the theory of quadrature domains [14, 15]. They also suggest that boundary identities may be transferred to conformal coordinates and studied through their Fourier coefficients.

A related computational approach begins with interior solutions of the parameter-dependent equation. Their boundary traces are then used to select the spectral parameter. In 1967, Fox, Henrici, and Moler implemented this idea with finite combinations

$$u_N(x;\lambda) = \sum_{j=1}^{N} c_j v_j(x;\lambda), \qquad (-\Delta - \lambda)v_j = 0,$$

and located candidate eigenvalues by seeking coefficients for which the boundary trace is small [16]. In 2005, Betcke and Trefethen observed that an unnormalized boundary residual may produce spurious candidates. They introduced interior normalization and formulated the test through subspace angles [17].

This boundary-residual viewpoint suggests an infinite-dimensional formulation: a family of Helmholtz solutions parametrized by boundary data may encode the Dirichlet condition through its boundary traces. Completeness requires this family to contain every Helmholtz solution in the relevant class. We pursue this idea below.

From this point onward, $\Omega \subset \mathbb{C}$ is bounded and simply connected, with $\partial\Omega$ a real-analytic Jordan curve. Let $\mathbb{D} \stackrel{\text{def}}{=} \{w \in \mathbb{C} : |w| < 1\}$. We fix $0 \in \Omega$ and let

$$\phi(w) = \sum_{j\geq 1} a_j w^j, \qquad \phi : \mathbb{D} \longrightarrow \Omega, \qquad \phi(0) = 0, \qquad \phi'(0) > 0,$$

be the normalized Riemann map. Let $d\omega_0$ be harmonic measure on $\partial\Omega$ with pole at zero, and set $\mathcal{H}_\Omega = L^2(\partial\Omega, d\omega_0)$. Section 2 collects the analytic and operator-theoretic tools used below.

We begin with boundary data $b \in \mathcal{H}_\Omega$. Its harmonic extension h_b is decomposed into holomorphic and antiholomorphic components, from which iterated primitive operators produce a Helmholtz solution $u_b(\cdot;\lambda)$. Its radial boundary value defines the Wronskian operator $W(\lambda) : \mathcal{H}_\Omega \to \mathcal{H}_\Omega$. Hence

$$W(\lambda)b = 0 \quad \Longleftrightarrow \quad u_b(\cdot;\lambda)\big|_{\partial\Omega} = 0.$$

The representation result in Section 2 supplies the converse: for real λ, every weak $H^1(\Omega)$-solution of the Helmholtz equation is generated by a unique boundary datum. Therefore $b \mapsto u_b(\cdot;\lambda)$ maps $\ker W(\lambda)$ linearly and bijectively onto the Dirichlet eigenspace.

Since $W(\lambda)$ acts on an infinite-dimensional Hilbert space, an ordinary determinant is unavailable. We prove in Section 3 that

$$W(\lambda) = I_{\mathcal{H}_\Omega} + K(\lambda), \qquad K(\lambda) \in \mathcal{S}_2(\mathcal{H}_\Omega),$$

Here $\mathcal{S}_2(\mathcal{H}_\Omega)$ denotes the Hilbert–Schmidt class. The 2-regularized determinant $\det{}_2 W(\lambda)$ is consequently well defined and entire. For every $\lambda > 0$,

$$\lambda \text{ is a Dirichlet eigenvalue of } \Omega \quad \Longleftrightarrow \quad \ker W(\lambda) \neq \{0\} \quad \Longleftrightarrow \quad \det{}_2 W(\lambda) = 0.$$

Moreover, $\dim\ker W(\lambda)$ equals the spectral multiplicity of λ; see Theorems 3.15 and 3.16.

In Section 4, we construct the Wronskian matrix from the Riemann mapping coefficients. Conformal pullback identifies $\mathcal{H}_\Omega$ with the standard Fourier space on $\partial\mathbb{D}$. In the corresponding conformal Fourier basis, the coefficients $\{a_j\}_{j\geq 1}$ determine the boundary multiplier and the primitive operators, and hence every entry of the matrix representing $W(\lambda)$. We also show that finite cutoffs of the Riemann mapping coefficients, Fourier modes, and operator series produce computable matrices. We further prove convergence of the corresponding regularized determinants for each fixed λ along a coordinated choice of truncation parameters; see Propositions 4.4 and 4.5.

In Section 5, we test the finite construction on the Neumann oval with $R = 1.8$. The forward experiment locates spectral candidates through the smallest singular values of the truncated Wronskian matrices and compares them across several truncation levels. A restricted coefficient-fitting experiment then fixes the remaining Riemann mapping coefficients and recovers a_1, a_3, a_5 from seven spectral candidates generated by the same finite model. These computations illustrate the forward evaluation of the Wronskian operator and its local use in parameter recovery.

These results turn the Riemann mapping coefficients into a computable spectral criterion: they determine an operator whose kernel recovers the Dirichlet eigenspaces and whose regularized determinant detects the eigenvalues. The remaining inverse questions are therefore explicit: they concern the injectivity and stability of the resulting coefficient-to-determinant map.

2 Preliminaries

We first collect the tools used to construct the Wronskian operator, prove that its Helmholtz parametrization is exhaustive, and define its regularized determinant. All Hilbert spaces are complex, and inner products are linear in the first argument.

2.1 Conformal coordinates and the Schwarz function

Let $\Omega \subset \mathbb{C}$ be a bounded simply connected domain whose boundary is a real-analytic Jordan curve. Translate a fixed interior point to zero, and let

$$\phi : \mathbb{D} \longrightarrow \Omega, \qquad \phi(0) = 0, \qquad \phi'(0) > 0, \qquad \psi = \phi^{-1}.$$

Write

$$\Omega^* = \{\overline{\eta} : \eta \in \Omega\}, \qquad d = \min_{\eta \in \partial\Omega} |\eta| > 0, \qquad L = \max_{\eta \in \partial\Omega} |\eta|.$$

For $f \in O(\Omega)$, define the reflected holomorphic function on Ω^* by

$$f^{\#}(\zeta) \stackrel{\text{def}}{=} \overline{f(\overline{\zeta})}.$$

Thus $f^{\#}(\overline{\eta}) = \overline{f(\eta)}$. In the context, we write $O(\Omega)$ and $C^{\omega}(\Omega)$ for the holomorphic and real-analytic functions, respectively.

For the Riemann map, we use the same reflection notation on the reflected disk:

$$\phi^{\#}(\zeta) \stackrel{\text{def}}{=} \overline{\phi(\overline{\zeta})}.$$

For $\eta = x + iy$, set

$$\partial_\eta = \tfrac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\overline{\eta}} = \tfrac{1}{2}(\partial_x + i\partial_y), \qquad \Delta = 4\partial_\eta \partial_{\overline{\eta}}.$$

Lemma 2.1. *The maps ϕ and ψ extend conformally across their boundaries. The Schwarz function S is holomorphic on a sufficiently small collar of $\partial\Omega$, satisfies $S(\eta) = \overline{\eta}$ on $\partial\Omega$, and obeys*

$$S(\phi(w)) = \phi^{\#}(1/w), \qquad \psi^{\#}(S(\phi(w))) = 1/w \tag{2.1}$$

on a sufficiently small annulus about $\partial\mathbb{D}$. Moreover, ϕ' is bounded and bounded away from zero on $\overline{\mathbb{D}}$.

Proof. Analytic boundary reflection extends the Riemann map and its inverse across the boundary [18, 14]. The local extensions agree on overlaps by analytic continuation. On $|w| = 1$, we have

$$S(\phi(w)) = \overline{\phi(w)} = \overline{\phi(\overline{1/w})} = \phi^{\#}(1/w).$$

The identity theorem extends this equality to a smaller annulus, and composition with $\psi^{\#}$ gives the second identity. The extended map is conformal, so continuity of $|\phi'|$ on the compact closed disk gives the final bounds. □

For a twice differentiable function v, the chain rule gives

$$\partial_w(v \circ \phi) = (\partial_\eta v) \circ \phi\, \phi', \qquad \partial_{\overline{w}}\partial_w(v \circ \phi) = |\phi'|^2(\partial_{\overline{\eta}}\partial_\eta v) \circ \phi.$$

Consequently,

$$\Delta_w(v \circ \phi) = |\phi'|^2(\Delta_\eta v) \circ \phi. \tag{2.2}$$

Thus conformal pullback preserves harmonicity. We use these coordinates for boundary data, while the Helmholtz construction remains in the physical variable η.

2.2 Fourier, Hardy, and boundary Sobolev spaces

We use normalized measure $d\theta/(2\pi)$ on the unit circle. Throughout, w denotes the variable in $\mathbb{D}$; on $\partial\mathbb{D}$ we write $w = e^{i\theta}$, and ρw, $0 \le \rho < 1$, denotes its radial contraction. For $v \in L^2(\partial\mathbb{D})$, put

$$\widehat{v}_j \stackrel{\text{def}}{=} \frac{1}{2\pi}\int_0^{2\pi} v(e^{i\theta})e^{-ij\theta}\, d\theta, \qquad j \in \mathbb{Z}.$$

By Fourier orthogonality,

$$v = \sum_{j\in\mathbb{Z}} \widehat{v}_j e^{ij\theta} \quad \text{in } L^2(\partial\mathbb{D}), \qquad \|v\|^2_{L^2(\partial\mathbb{D})} = \sum_{j\in\mathbb{Z}} |\widehat{v}_j|^2.$$

For $s \ge 0$, set

$$\|v\|^2_{H^s(\partial\mathbb{D})} \stackrel{\text{def}}{=} \sum_{j\in\mathbb{Z}} (1 + j^2)^s |\widehat{v}_j|^2.$$

Define harmonic measure at zero by

$$\int_{\partial\Omega} b\, d\omega_0 \stackrel{\text{def}}{=} \frac{1}{2\pi}\int_0^{2\pi} b(\phi(e^{i\theta}))\, d\theta.$$

Equivalently, $d\omega_0 = ds/(2\pi|\phi' \circ \psi|)$. Define

$$\mathcal{H}_\Omega \stackrel{\text{def}}{=} L^2(\partial\Omega, d\omega_0), \qquad C_\phi b \stackrel{\text{def}}{=} b \circ \phi. \tag{2.3}$$

Lemma 2.2. *The map $C_\phi : \mathcal{H}_\Omega \to L^2(\partial\mathbb{D})$ is unitary, with*

$$C_\phi^{-1}F = F \circ \psi, \qquad \|C_\phi\| = \|C_\phi^{-1}\| = 1.$$

Proof. By the definition of harmonic measure,

$$\|C_\phi b\|^2_{L^2(\partial\mathbb{D})} = \frac{1}{2\pi}\int_0^{2\pi} |b(\phi(e^{i\theta}))|^2\, d\theta = \|b\|^2_{\mathcal{H}_\Omega}.$$

Thus C_ϕ is an isometry, while $F \mapsto F \circ \psi$ gives a preimage of every $F \in L^2(\partial\mathbb{D})$. □

Since C_ϕ is unitary, transporting the Fourier system $(e^{ij\theta})_{j\in\mathbb{Z}}$ to $\mathcal{H}_\Omega$ gives the orthonormal basis $\widetilde{B} = (E_j)_{j\in\mathbb{Z}}$, where

$$E_j(\phi(e^{i\theta})) = e^{ij\theta}, \qquad E_j(\eta) = \begin{cases} \psi(\eta)^j, & j \ge 0, \\ (\psi^{\#}(S(\eta)))^{-j}, & j < 0. \end{cases} \tag{2.4}$$

The second expression is defined on a boundary collar, and (2.1) identifies its pullback with w^j. In particular,

$$\langle E_j, E_k\rangle_{\mathcal{H}_\Omega} = \langle e^{ij\theta}, e^{ik\theta}\rangle_{L^2(\partial\mathbb{D})} = \delta_{jk}.$$

For $b \in \mathcal{H}_\Omega$, write $\widehat{b}_j = \langle b, E_j\rangle_{\mathcal{H}_\Omega}$. For $s \geq 0$, we transport the boundary Sobolev spaces through the conformal parametrization by setting

$$H^s(\partial\Omega) \stackrel{\text{def}}{=} \{b \in \mathcal{H}_\Omega : C_\phi b \in H^s(\partial\mathbb{D})\}, \qquad \|b\|_{H^s(\partial\Omega)} \stackrel{\text{def}}{=} \|C_\phi b\|_{H^s(\partial\mathbb{D})}.$$

These norms are equivalent to the usual boundary Sobolev norms defined in local arclength coordinates.

The disk Hardy space and its conformal counterpart are

$$H^2(\mathbb{D}) = \left\{F(w) = \sum_{k\geq 0} a_k w^k : \sum_{k\geq 0} |a_k|^2 < \infty\right\},$$

$$H^2_{\text{hol}}(\Omega) \stackrel{\text{def}}{=} \{f \in O(\Omega) : f \circ \phi \in H^2(\mathbb{D})\}, \qquad \|f\|_{H^2_{\text{hol}}(\Omega)} \stackrel{\text{def}}{=} \|f \circ \phi\|_{H^2(\mathbb{D})}.$$

The notation $H^s(\Omega)$ always denotes a Sobolev space, and Hardy boundary values are radial L^2 limits.

Lemma 2.3. *If $F(w) = \sum_{k\geq 0} a_k w^k \in H^2(\mathbb{D})$, then*

$$|F(w)| \leq \frac{\|F\|_{H^2(\mathbb{D})}}{\sqrt{1-|w|^2}}.$$

If $F' \in H^2(\mathbb{D})$, its boundary value belongs to $H^1(\partial\mathbb{D})$, with

$$\|F\|^2_{H^1(\partial\mathbb{D})} = \|F\|^2_{H^2(\mathbb{D})} + \|F'\|^2_{H^2(\mathbb{D})}. \tag{2.5}$$

Moreover,

$$\int_{\mathbb{D}} |F|^2\, dA = \pi \sum_{k\geq 0} \frac{|a_k|^2}{k+1}, \qquad \int_{\mathbb{D}} |\nabla F|^2\, dA = 2\pi \sum_{k\geq 1} k|a_k|^2. \tag{2.6}$$

The right-hand sides may be infinite.

Proof. Cauchy–Schwarz and the geometric series give

$$|F(w)| \leq \left(\sum_{k\geq 0} |a_k|^2\right)^{1/2} \left(\sum_{k\geq 0} |w|^{2k}\right)^{1/2} = \frac{\|F\|_{H^2}}{\sqrt{1-|w|^2}}.$$

The derivative series gives $\|F'\|^2_{H^2} = \sum_{k\geq 1} k^2|a_k|^2$, proving the boundary identity. Fourier orthogonality followed by radial integration gives

$$\int_{\mathbb{D}} |F|^2 dA = 2\pi \sum_{k\geq 0} |a_k|^2 \int_0^1 \rho^{2k+1} d\rho = \pi \sum_{k\geq 0} \frac{|a_k|^2}{k+1}.$$

Since $|\nabla F|^2 = 2|F'|^2$, applying the same identity to F' gives the second area formula. Monotone convergence as $\rho \to 1^-$ covers the general case. □

2.3 Harmonic extension and Sobolev traces

Lemma 2.4. *For $(b_j)_{j\in\mathbb{Z}} \in \ell^2(\mathbb{Z})$, the series*

$$v(\rho e^{i\theta}) \stackrel{\text{def}}{=} \sum_{j\in\mathbb{Z}} b_j \rho^{|j|} e^{ij\theta}$$

defines a harmonic function on $\mathbb{D}$, converges with all derivatives on compact subsets, and has radial L^2 boundary value $\sum_j b_j e^{ij\theta}$. Moreover,

$$\int_{\mathbb{D}} |v|^2 \, dA = \pi \sum_{j\in\mathbb{Z}} \frac{|b_j|^2}{|j|+1}, \qquad \int_{\mathbb{D}} |\nabla v|^2 \, dA = 2\pi \sum_{j\in\mathbb{Z}} |j| \, |b_j|^2. \tag{2.7}$$

Thus boundary data in $H^{1/2}(\partial\mathbb{D})$ have harmonic extensions in $H^1(\mathbb{D})$.

Proof. On every closed subdisk, Cauchy–Schwarz applied after any number of derivatives gives normal convergence. Each summand is harmonic, and Fourier orthogonality gives

$$\left\| v(\rho e^{i\theta}) - \sum_j b_j e^{ij\theta} \right\|_{L^2}^2 = \sum_j |b_j|^2 (1-\rho^{|j|})^2 \longrightarrow 0.$$

Integrating the orthogonal modes yields

$$\int_{\mathbb{D}} |v|^2 dA = 2\pi \sum_j |b_j|^2 \int_0^1 \rho^{2|j|+1} d\rho,$$

and

$$\int_{\mathbb{D}} |\nabla v|^2 dA = 4\pi \sum_{j\neq 0} |j|^2 |b_j|^2 \int_0^1 \rho^{2|j|-1} d\rho.$$

These are the two identities in (2.7). Their sum is bounded by $3\pi \sum_j (1+j^2)^{1/2} |b_j|^2$, proving the last assertion. □

Lemma 2.5. *The Sobolev trace is bounded,*

$$\gamma : H^1(\Omega) \longrightarrow H^{1/2}(\partial\Omega), \qquad \ker \gamma = H_0^1(\Omega).$$

There are constants $C_{\text{tr}}, C_{\text{ext}} > 0$, depending only on the fixed domain and conformal parametrization, such that

$$\|\gamma v\|_{H^{1/2}(\partial\Omega)} \leq C_{\text{tr}} \|v\|_{H^1(\Omega)}$$

and the conformally transported harmonic extension h of $b \in H^{1/2}(\partial\Omega)$ satisfies

$$\|h\|_{H^1(\Omega)} \leq C_{\text{ext}} \|b\|_{H^{1/2}(\partial\Omega)}.$$

Every $f \in O(\Omega) \cap H^1(\Omega)$ belongs to $H^2_{\text{hol}}(\Omega)$, and its Hardy boundary value equals its Sobolev trace.

Proof. The trace statements are standard on smooth bounded domains [19]. If $v = h \circ \phi$, change of variables and conformal invariance of the Dirichlet integral give

$$\int_\Omega |h|^2 dA \leq \|\phi'\|_\infty^2 \int_{\mathbb{D}} |v|^2 dA, \qquad \int_\Omega |\nabla h|^2 dA = \int_{\mathbb{D}} |\nabla v|^2 dA.$$

Equation (2.7) therefore gives the extension estimate, for example with $C_{\mathrm{ext}} = \sqrt{\pi(\|\phi'\|_\infty^2 + 2)}$.

For the last assertion, write $F = f \circ \phi = \sum_{k\geq 0} a_k w^k$. Smooth coordinate changes preserve H^1, and (2.6) gives

$$\sum_{k\geq 0} \frac{|a_k|^2}{k+1} + \sum_{k\geq 1} k|a_k|^2 < \infty.$$

Hence $\sum_k |a_k|^2 < \infty$. The Taylor polynomials converge both in $H^1(\mathbb{D})$ and in $H^2(\mathbb{D})$, so uniqueness of the L^2 limit identifies the Sobolev and Hardy boundary values. □

2.4 Analytic regularity and Vekua representation

We next record the representation result needed to show that the Helmholtz family constructed in Section 3 omits no weak solution. The Bergman–Vekua theory represents solutions of a planar elliptic equation by holomorphic data. For the Helmholtz equation, explicit Vekua transforms between harmonic and Helmholtz solutions, with Sobolev continuity on star-shaped Lipschitz domains, are developed in [20]. We use the global representation on a bounded simply connected domain and the Sobolev control proved in [21, 22].

Lemma 2.6. *Let $\lambda \in \mathbb{C}$. A weak solution $u \in H^1(\Omega)$ of $-\Delta u = \lambda u$ has a real-analytic representative in Ω. Given $K \Subset \Omega$ and $m \in \mathbb{N}_0$, there is a constant $C = C(\Omega, K, m, \lambda) > 0$ for which*

$$\|u\|_{C^m(K)} \leq C\|u\|_{H^1(\Omega)}.$$

In particular, evaluation at zero is continuous on this solution space.

Proof. Choose an open set U with $K \Subset U \Subset \Omega$. Iterating the interior estimate for $-\Delta u = \lambda u$ on a finite chain of open sets between K and U gives, for every integer $s \geq 0$,

$$\|u\|_{H^{s+2}(K)} \leq C_{K,U,s,\lambda}\|u\|_{L^2(U)} \leq C_{K,U,s,\lambda}\|u\|_{H^1(\Omega)}.$$

Taking $s = m$ and applying the two-dimensional Sobolev embedding $H^{m+2}(K) \hookrightarrow C^m(K)$ proves the displayed estimate [19]. Since the coefficients of $-\Delta - \lambda$ are analytic, analytic elliptic regularity gives a real-analytic representative [23]. For nonreal λ, write $u = u_1 + iu_2$. The scalar equation then becomes a two-component elliptic system for (u_1, u_2). Its principal operator is the diagonal Laplacian and its lower-order coefficients are constant, so analytic elliptic regularity applies to both components. □

To state the result, we treat ζ as an independent complexification of $\overline{\eta}$; it therefore ranges over Ω^*. Since $\Delta = 4\partial_\eta \partial_{\overline{\eta}}$, the complexified Helmholtz equation contains no first-order derivatives and takes the form $U_{\eta\zeta} = -(\lambda/4)U$. Its characteristic sets through the origin are the two coordinate axes $\zeta = 0$ and $\eta = 0$. The values of U on these axes are the holomorphic data used below.

Theorem 2.7. *Let $\lambda \in \mathbb{R}$, and let $u \in H^1(\Omega)$ satisfy $-\Delta u = \lambda u$ weakly. There is a unique $U \in \mathcal{O}(\Omega \times \Omega^*)$ such that*

$$U(\eta, \overline{\eta}) = u(\eta), \qquad U_{\eta\zeta} = -\frac{\lambda}{4}U. \tag{2.8}$$

Its normalized axis data

$$f(\eta) \stackrel{\mathrm{def}}{=} U(\eta, 0), \qquad g^{\#}(\zeta) \stackrel{\mathrm{def}}{=} U(0, \zeta) - U(0, 0) \tag{2.9}$$

satisfy $g(0) = 0$, $f, g \in H^1(\Omega)$, *and*

$$\|f\|_{H^1(\Omega)} + \|g\|_{H^1(\Omega)} \le C_{\Omega,\lambda}\|u\|_{H^1(\Omega)}$$

for some $C_{\Omega,\lambda} > 0$ *independent of* u.

Proof. Assume first that u is real-valued. By Lemma 2.6, we use its real-analytic representative. Under the change of variables $\eta = x + iy$, $\zeta = x - iy$, the equation becomes

$$\left(\partial_\eta \partial_\zeta + \frac{\lambda}{4}\right) U = 0.$$

This is the constant-coefficient case of the complexified elliptic equations treated by Bergman–Vekua theory. The corresponding continuation and representation results give a holomorphic function U on $\Omega \times \Omega^*$ and parametrize the real solution by the normalized holomorphic datum

$$v(\eta) \stackrel{\text{def}}{=} 2U(\eta, 0) - u(0), \qquad \|v\|_{H^1(\Omega)} \le C_{\mathrm{V}}\|u\|_{H^1(\Omega)}$$

for some $C_{\mathrm{V}} > 0$ independent of u; see [21, 22, 20]. To relate this representation to (2.9), write locally $U(\eta, \zeta) = \sum_{j,k\ge 0} c_{jk}\eta^j\zeta^k$. The equation gives

$$(j + 1)(k + 1)c_{j+1,k+1} = -\frac{\lambda}{4}c_{jk}.$$

Thus the coefficients on the two axes, which share the value c_{00}, determine all coefficients with positive indices. Consequently $U(\eta, 0)$ and $U(0, \zeta)$ are precisely the two axis data used in (2.9).

To verify that the physical diagonal determines the complexification, expand the difference of two extensions near $(0, 0)$ as $\sum_{i,j\ge 0} c_{ij}\eta^i\zeta^j$. Its diagonal restriction is zero, and hence

$$0 = \partial_\eta^i \partial_{\overline{\eta}}^j \sum_{a,b\ge 0} c_{ab}\eta^a\overline{\eta}^b\Big|_{\eta=0} = i!\,j!\,c_{ij}.$$

The extensions agree near the origin. The identity theorem first continues equality in η and then in ζ, so they agree on $\Omega \times \Omega^*$.

For real u, uniqueness applied to $(\eta, \zeta) \mapsto \overline{U(\overline{\zeta}, \overline{\eta})}$ gives

$$U(\eta, \zeta) = \overline{U(\overline{\zeta}, \overline{\eta})}.$$

The axis data therefore satisfy

$$f = \tfrac{1}{2}(v + u(0)), \qquad g = \tfrac{1}{2}(v - u(0)), \qquad g(0) = 0.$$

The point-evaluation bound in Lemma 2.6 and the estimate for v give the stated H^1 bound.

For complex-valued $u = u_1 + iu_2$, apply the real result to each part and set

$$U \stackrel{\text{def}}{=} U_1 + iU_2, \qquad f \stackrel{\text{def}}{=} f_1 + if_2, \qquad g \stackrel{\text{def}}{=} g_1 - ig_2.$$

Then $g^\# = g_1^\# + ig_2^\#$, so the equations and axis identities remain valid. The triangle inequality gives the required bound for some $C_{\Omega,\lambda} > 0$, and uniqueness of U fixes both axis data. □

2.5 Operator ideals and regularized determinants

We use the standard Schatten operator ideals and regularized determinants described in [24]. For Banach spaces X and Y, let $\mathcal{B}(X,Y)$ denote the bounded linear operators from X to Y, and set $\mathcal{B}(X) \stackrel{\text{def}}{=} \mathcal{B}(X,X)$. Let $\mathcal{X},\mathcal{Y}$ be Hilbert spaces. The Hilbert–Schmidt class $\mathcal{S}_2(\mathcal{X},\mathcal{Y})$ consists of those bounded operators $B:\mathcal{X}\to\mathcal{Y}$ for which

$$\|B\|_{\mathcal{S}_2}^2 \stackrel{\text{def}}{=} \sum_j \|Be_j\|_{\mathcal{Y}}^2 < \infty$$

for one, and hence every, orthonormal basis (e_j) of $\mathcal{X}$. Such operators are compact. The trace class $\mathcal{S}_1(\mathcal{X})$ consists of compact operators on $\mathcal{X}$ whose singular values are summable; equivalently, $B\in\mathcal{S}_1$ when $\operatorname{tr}|B|<\infty$. We abbreviate $\mathcal{S}_p(\mathcal{X},\mathcal{X})$ to $\mathcal{S}_p(\mathcal{X})$.

Lemma 2.8. *Let $B\in\mathcal{S}_2(\mathcal{X},\mathcal{Y})$, $A\in\mathcal{B}(\mathcal{Y},\mathcal{Z})$, and $C\in\mathcal{B}(\mathcal{W},\mathcal{X})$. Then $ABC\in\mathcal{S}_2(\mathcal{W},\mathcal{Z})$ and*

$$\|ABC\|_{\mathcal{S}_2} \le \|A\|\,\|B\|_{\mathcal{S}_2}\,\|C\|.$$

Moreover, $\|B\|\le\|B\|_{\mathcal{S}_2}$, the ideal $\mathcal{S}_2$ is complete, and a locally uniform limit of analytic $\mathcal{S}_2$-valued functions is analytic.

Proof. Expanding in an orthonormal basis of the output space gives $\sum_j\|Be_j\|^2=\sum_k\|B^*f_k\|^2$, which proves basis independence. The estimate $\sum_j\|ABe_j\|^2\le\|A\|^2\sum_j\|Be_j\|^2$, applied also to C^*B^*, gives the ideal inequality. The remaining assertions are the standard completeness and analytic-limit properties of the Hilbert–Schmidt ideal [24]. □

Lemma 2.9. *For $K\in\mathcal{S}_2(\mathcal{H}_\Omega)$, the operator $(I+K)e^{-K}-I$ belongs to $\mathcal{S}_1(\mathcal{H}_\Omega)$. The 2-regularized determinant is*

$$\det{}_2(I+K) \stackrel{\text{def}}{=} \det\big((I+K)e^{-K}\big) = \prod_j (1+\mu_j)e^{-\mu_j}, \tag{2.10}$$

where the nonzero eigenvalues μ_j of K are counted with algebraic multiplicity. The product is convergent, so $\det_2(I+K)$ is well defined. It is analytic on analytic $\mathcal{S}_2$-valued families and continuous in the $\mathcal{S}_2$ norm. Furthermore,

$$\det{}_2(I+K)=0 \iff I+K \text{ is non-invertible} \iff \ker(I+K)\neq\{0\}.$$

Proof. Since $K^2\in\mathcal{S}_1$, cancellation of the linear term gives

$$(I+K)e^{-K}-I=\sum_{m=2}^{\infty}\frac{(-1)^{m-1}(m-1)}{m!}K^m.$$

The trace-ideal product estimate yields

$$\|(I+K)e^{-K}-I\|_{\mathcal{S}_1}\le\|K\|_{\mathcal{S}_2}^2\sum_{m=2}^{\infty}\frac{\|K\|^{m-2}}{(m-2)!}=\|K\|_{\mathcal{S}_2}^2e^{\|K\|}.$$

The regularized determinant theorem gives the product formula, analyticity, and continuity [24]. Since e^{-K} is invertible, the determinant vanishes exactly when $I+K$ is non-invertible. Finally, for compact K, injectivity of $I+K$ implies surjectivity; the bounded inverse theorem gives the last equivalence. □

3 Wronskian Operator

In this section, we construct a boundary operator for (1.1). We show that a boundary datum b determines a harmonic seed h_b, from which an integral series produces a Helmholtz solution $u_b(\cdot;\lambda)$. We further define $W(\lambda)b$ as the radial boundary trace of $u_b(\cdot;\lambda)$, and prove that $b \mapsto u_b$ maps $\ker W(\lambda)$ bijectively onto the Dirichlet eigenspace.

3.1 Harmonic data and primitive operators

Let $b \in \mathcal{H}_\Omega$, with $\widehat{b}_j = \langle b, E_j\rangle_{\mathcal{H}_\Omega}$. In (3.1), the point $\rho e^{i\theta}$ approaches the boundary radially. The variable $w \in \mathbb{D}$ in the subsequent holomorphic splitting is independent of ρ. Define

$$h_b(\phi(\rho e^{i\theta})) \overset{\text{def}}{=} \sum_{j\in\mathbb{Z}} \widehat{b}_j \rho^{|j|} e^{ij\theta}, \qquad 0 \le \rho < 1, \tag{3.1}$$

and separate its positive and negative frequencies by

$$f_b(\phi(w)) \overset{\text{def}}{=} \sum_{j\ge 0} \widehat{b}_j w^j, \qquad g_b(\phi(w)) \overset{\text{def}}{=} \sum_{j\ge 1} \overline{\widehat{b}_{-j}} w^j.$$

Then

$$h_b = f_b + \overline{g_b}, \qquad f_b, g_b \in H^2_{\text{hol}}(\Omega), \qquad g_b(0) = 0. \tag{3.2}$$

Lemma 3.1. *The function h_b is the unique harmonic extension of b with bounded radial L^2 means, and the splitting* (3.2) *is unique. Moreover,*

$$\|f_b\|^2_{H^2_{\text{hol}}} + \|g_b\|^2_{H^2_{\text{hol}}} = \|b\|^2_{\mathcal{H}_\Omega}.$$

If $b \in H^{1/2}(\partial\Omega)$, then

$$\|h_b\|_{H^1(\Omega)} + \|f_b\|_{H^1(\Omega)} + \|g_b\|_{H^1(\Omega)} \le (1+\sqrt{2})C_{\text{ext}}\|b\|_{H^{1/2}(\partial\Omega)}.$$

Proof. Lemma 2.4 and (2.2) prove harmonicity and radial convergence. The orthonormality of $(E_j)_{j\in\mathbb{Z}}$ gives

$$\|f_b\|^2_{H^2_{\text{hol}}} = \sum_{j\ge 0} |\widehat{b}_j|^2, \qquad \|g_b\|^2_{H^2_{\text{hol}}} = \sum_{j<0} |\widehat{b}_j|^2.$$

The boundary coefficients determine the regular radial factor $\rho^{|j|}$ of every harmonic Fourier mode, proving uniqueness. The two frequency projections are orthogonal in $H^{1/2}(\partial\Omega)$. Hence, if their boundary values are denoted by b_+ and b_-, then

$$\|b_+\|^2_{H^{1/2}} + \|b_-\|^2_{H^{1/2}} = \|b\|^2_{H^{1/2}}, \qquad \|b_+\|_{H^{1/2}} + \|b_-\|_{H^{1/2}} \le \sqrt{2}\|b\|_{H^{1/2}}.$$

Applying Lemma 2.5 to these projections and to b gives the final estimate. □

However, the harmonic seed and its Helmholtz image need not have the same boundary trace. To construct the latter, we define the primitive operator

$$T_\Omega : H^2_{\text{hol}}(\Omega) \longrightarrow H^2_{\text{hol}}(\Omega), \qquad (T_\Omega f)(\eta) \overset{\text{def}}{=} \int_0^\eta f(\zeta)\, d\zeta, \qquad T^0_\Omega = I. \tag{3.3}$$

The integral is path independent because Ω is simply connected.

Proposition 3.2. *The operator T_Ω is Hilbert–Schmidt. For $r \geq 1$,*

$$\|T_\Omega\| \leq \|\phi'\|_\infty, \qquad \|T_\Omega^r\|_{\mathcal{S}_2} \leq \frac{\pi}{\sqrt{6}}\|\phi'\|_\infty^r. \tag{3.4}$$

Every $T_\Omega^r f$ vanishes to order at least r at zero, and

$$T_\Omega^r(\eta^k) = \frac{k!}{(k+r)!}\eta^{k+r}, \qquad k \geq 0. \tag{3.5}$$

Furthermore,

$$\|(T_\Omega^r f)\circ\phi\|_{H^1(\partial\mathbb{D})} \leq \sqrt{2}\|\phi'\|_\infty^r \|f\|_{H^2_{\mathrm{hol}}(\Omega)}. \tag{3.6}$$

Proof. For $F = f\circ\phi$, let $(T_{\mathbb{D}}F)(w) \overset{\text{def}}{=} \int_0^w F(\zeta)\,d\zeta$. Changing variables in the path integral gives

$$(T_\Omega f)(\phi(w)) = \int_0^w \phi'(\zeta)F(\zeta)\,d\zeta = T_{\mathbb{D}}(\phi' F)(w).$$

Multiplication by ϕ' has norm at most $\|\phi'\|_\infty$, while $T_{\mathbb{D}}(w^k) = w^{k+1}/(k+1)$. Hence

$$\|T_{\mathbb{D}}\|_{\mathcal{S}_2}^2 = \sum_{k\geq 0}\left\|\frac{w^{k+1}}{k+1}\right\|_{H^2}^2 = \sum_{k\geq 0}\frac{1}{(k+1)^2} = \frac{\pi^2}{6}.$$

The ideal property gives (3.4). Successive integration raises the order of vanishing and gives (3.5). Finally, set $F_r = (T_\Omega^r f)\circ\phi$. Since $F_r' = \phi' F_{r-1}$, induction gives

$$\|F_r\|_{H^2(\mathbb{D})} \leq \|\phi'\|_\infty^r\|f\|_{H^2_{\mathrm{hol}}(\Omega)}, \qquad \|F_r'\|_{H^2(\mathbb{D})} \leq \|\phi'\|_\infty^r\|f\|_{H^2_{\mathrm{hol}}(\Omega)}.$$

Therefore (2.5) gives

$$\|F_r\|_{H^1(\partial\mathbb{D})}^2 = \|F_r\|_{H^2(\mathbb{D})}^2 + \|F_r'\|_{H^2(\mathbb{D})}^2 \leq 2\|\phi'\|_\infty^{2r}\|f\|_{H^2_{\mathrm{hol}}(\Omega)}^2,$$

which is (3.6). □

3.2 Construction and completeness of the Helmholtz solutions

We first verify that the primitive series gives a well-defined Helmholtz solution for every boundary datum. Its surjectivity onto the weak solution space will then follow from the Vekua representation.

For $r \geq 1$, holomorphicity and $(T_\Omega^r f)' = T_\Omega^{r-1} f$ give

$$\partial_\eta\partial_{\overline{\eta}}\left(\overline{\eta}^r T_\Omega^r f(\eta)\right) = r\overline{\eta}^{r-1}T_\Omega^{r-1}f(\eta).$$

The factorial in the following series cancels this factor. We define

$$u_b(\eta;\lambda) \overset{\text{def}}{=} \sum_{r=0}^{\infty}\frac{(-\lambda/4)^r}{r!}\left(\overline{\eta}^r T_\Omega^r f_b(\eta) + \eta^r\overline{T_\Omega^r g_b(\eta)}\right), \qquad \lambda\in\mathbb{C}. \tag{3.7}$$

Only the geometric coefficients are conjugated in the second term; the parameter λ is not.

Proposition 3.3. *For every $b \in \mathcal{H}_\Omega$ and $\lambda \in \mathbb{C}$, the series* (3.7) *converges with all derivatives on compact subsets of Ω, locally uniformly in λ. Its sum is real analytic, depends complex linearly on b, and satisfies*

$$-\Delta u_b = \lambda u_b \quad \text{in } \Omega.$$

It has a radial $\mathcal{H}_\Omega$ boundary value. Moreover, $u_b(\cdot;0) = h_b$, and $u_b \equiv 0$ implies $b = 0$.

Proof. Introduce the holomorphic complexification

$$U_b(\eta,\zeta;\lambda) \stackrel{\text{def}}{=} \sum_{r\geq 0} \frac{(-\lambda/4)^r}{r!}\left(\zeta^r T_\Omega^r f_b(\eta) + \eta^r (T_\Omega^r g_b)^{\#}(\zeta)\right).$$

For compact $K_1 \Subset \Omega$, $K_2 \Subset \Omega^*$, the Hardy pointwise bound and Proposition 3.2 yield a constant $C_{K_1,K_2} > 0$, independent of r, b, λ, such that

$$\left|\zeta^r T_\Omega^r f_b(\eta) + \eta^r (T_\Omega^r g_b)^{\#}(\zeta)\right| \leq C_{K_1,K_2}(L\|\phi'\|_\infty)^r \|b\|_{\mathcal{H}_\Omega}.$$

For $|\lambda| \leq \Lambda$, the weighted majorant sums to

$$C_{K_1,K_2} e^{\Lambda L\|\phi'\|_\infty/4}\|b\|_{\mathcal{H}_\Omega}.$$

The series is therefore normal on compact subsets of the product. Cauchy estimates on a slightly larger compact set give the same conclusion for every derivative series.

Termwise differentiation and an index shift give

$$\partial_\eta \partial_{\overline{\eta}} \sum_{r\geq 0} \frac{(-\lambda/4)^r}{r!}\overline{\eta}^r T_\Omega^r f_b = -\frac{\lambda}{4}\sum_{r\geq 0}\frac{(-\lambda/4)^r}{r!}\overline{\eta}^r T_\Omega^r f_b.$$

The reflected part satisfies the same identity. Since $\Delta = 4\partial_\eta\partial_{\overline{\eta}}$, the restriction $u_b(\eta;\lambda) = U_b(\eta,\overline{\eta};\lambda)$ solves the Helmholtz equation.

It remains to identify the boundary value. Denote by $V_{r,\rho}$ the r-th bracket in (3.7), pulled back to $\eta = \phi(\rho e^{i\theta})$. It has an L^2 radial limit and obeys

$$\sup_{0<\rho<1}\|V_{r,\rho}\|_{L^2(\partial\mathbb{D})} \leq \sqrt{2}(L\|\phi'\|_\infty)^r\|b\|_{\mathcal{H}_\Omega}.$$

The factorially weighted majorant is summable, locally uniformly in λ. Dominated convergence in $L^2(\partial\mathbb{D})$ therefore permits passage of the radial limit through the series. At $\lambda = 0$, only the term $f_b + \overline{g_b} = h_b$ remains.

Every correction of positive order contains positive powers of both complex variables near zero. Consequently,

$$\partial_\eta^k u_b(0;\lambda) = f_b^{(k)}(0) \quad (k \geq 0), \qquad \partial_{\overline{\eta}}^k u_b(0;\lambda) = \overline{g_b^{(k)}(0)} \quad (k \geq 1).$$

If $u_b = 0$, both holomorphic data vanish by analytic continuation, so $b = 0$. Complex linearity follows from the Fourier splitting and the final conjugation in the reflected term. □

Definition 3.4. For $\lambda \in \mathbb{C}$, define the boundary Wronskian $W(\lambda) : \mathcal{H}_\Omega \to \mathcal{H}_\Omega$ by

$$W(\lambda)b \stackrel{\text{def}}{=} C_\phi^{-1}\left(\lim_{\rho\to 1^-} u_b(\phi(\rho e^{i\theta});\lambda)\right), \tag{3.8}$$

where the limit is taken in $L^2(\partial\mathbb{D})$.

The preceding estimate shows that $W(\lambda)$ is bounded and complex linear, with $W(0) = I$. Its kernel consists of precisely those harmonic seeds whose Helmholtz images have zero radial boundary trace.

Remark 3.5. In the conformal Fourier basis, the columns of $W(\lambda)$ are the boundary traces of the Helmholtz solutions generated by the basis elements. A nonzero kernel therefore records a linear dependence among these traces, in analogy with the classical Wronskian criterion. The boundary-recurrence origin of this terminology is recalled in the Appendix.

It remains to show that this parametrization contains every weak Helmholtz solution in the stated Sobolev class.

Theorem 3.6. *For real λ, every weak $H^1(\Omega)$ solution of $-\Delta u = \lambda u$ equals u_b for a unique $b \in H^{1/2}(\partial\Omega)$.*

Proof. Let U, f, g be supplied by Theorem 2.7. Put $h = f + \overline{g}$ and $b = \gamma h$. Then $b \in H^{1/2}(\partial\Omega)$, and uniqueness of the harmonic extension with the normalization $g(0) = 0$ identifies $f = f_b$ and $g = g_b$.

Let U_b be the complexified series used above. Since all positive integration orders vanish at zero,

$$U_b(\eta, 0) = f(\eta), \qquad U_b(0, \zeta) = f(0) + g^{\#}(\zeta).$$

Thus U and U_b have the same axis data and both satisfy $U_{\eta\zeta} = -\lambda U/4$. If $U - U_b = \sum_{i,j\geq 0} \delta_{ij}\eta^i\zeta^j$ near the origin, the equal axis data give $\delta_{i0} = \delta_{0j} = 0$, while coefficient comparison gives

$$(i+1)(j+1)\delta_{i+1,j+1} = -\frac{\lambda}{4}\delta_{ij}. \tag{3.9}$$

Repeated application reduces every δ_{ij} to an axis coefficient, so the difference vanishes near $(0, 0)$. The one-variable identity theorem, applied first in η and then in ζ, extends this equality to $\Omega \times \Omega^*$. Restriction to $\zeta = \overline{\eta}$ gives $u = u_b$. Uniqueness of b follows from Proposition 3.3. □

The construction on harmonic monomials produces regular Fourier–Bessel waves. Define

$$\mathfrak{B}_k(x) \overset{\text{def}}{=} \sum_{m=0}^{\infty} \frac{x^m}{m!(k+m)!}, \qquad k \in \mathbb{N}_0. \tag{3.10}$$

Lemma 3.7. *The functions $\mathfrak{B}_k$ are entire and satisfy*

$$\mathfrak{B}_k^{(n)} = \mathfrak{B}_{k+n}, \qquad J_k(t) = (t/2)^k \mathfrak{B}_k(-t^2/4).$$

Here J_k is the Bessel function of the first kind, $k, n \in \mathbb{N}_0$, and $t \in \mathbb{C}$. For every $\lambda \in \mathbb{C}$, the seeds $\eta^k/k!$ and $\overline{\eta}^k/k!$ generate $\eta^k \mathfrak{B}_k(-\lambda|\eta|^2/4)$ and $\overline{\eta}^k \mathfrak{B}_k(-\lambda|\eta|^2/4)$, respectively.

Proof. The defining power series converges everywhere, and termwise differentiation gives $\mathfrak{B}_k' = \mathfrak{B}_{k+1}$. The Bessel identity follows from the power series of J_k. Finally, (3.5) gives

$$\sum_{r\geq 0} \frac{(-\lambda/4)^r}{r!} \overline{\eta}^r T_\Omega^r \left(\frac{\eta^k}{k!}\right) = \eta^k \sum_{r\geq 0} \frac{(-\lambda|\eta|^2/4)^r}{r!(k+r)!},$$

which is the first formula; the reflected calculation is identical. □

We write these solutions as

$$\mathcal{U}_p(\eta,\overline{\eta};\lambda) \stackrel{\text{def}}{=} \begin{cases} \eta^p \mathfrak{B}_p(-\lambda|\eta|^2/4), & p \geq 0, \\ \overline{\eta}^{-p} \mathfrak{B}_{-p}(-\lambda|\eta|^2/4), & p < 0. \end{cases} \tag{3.11}$$

For real $\lambda > 0$ and $\eta = \varrho e^{i\vartheta}$,

$$\mathcal{U}_p = \left(\frac{2}{\sqrt{\lambda}}\right)^{|p|} J_{|p|}(\sqrt{\lambda}\,\varrho) e^{ip\vartheta}.$$

3.3 Schwarz multiplication and boundary coefficient operators

On the boundary, $S(\eta) = \overline{\eta}$. Let H be the multiplication operator

$$(Hb)(\eta) \stackrel{\text{def}}{=} \eta S(\eta) b(\eta) = |\eta|^2 b(\eta), \tag{3.12}$$

and, in conformal coordinates,

$$q(w) \stackrel{\text{def}}{=} \phi(w)\phi^{\#}(1/w), \qquad \widehat{q}_\ell \stackrel{\text{def}}{=} \frac{1}{2\pi}\int_0^{2\pi} q(e^{i\theta}) e^{-i\ell\theta}\, d\theta. \tag{3.13}$$

Thus $q(e^{i\theta}) = |\phi(e^{i\theta})|^2$.

Proposition 3.8. *The multiplication operator H is positive and boundedly invertible on $\mathcal{H}_\Omega$, with*

$$\|H\| = L^2, \qquad \|H^{-1}\| = d^{-2}, \qquad d^2 \leq \widehat{q}_0 \leq L^2. \tag{3.14}$$

Its matrix $M_H = [H]_{\widetilde{B}}$ is the Toeplitz matrix

$$(M_H)_{k,j} = \widehat{q}_{k-j}, \qquad k, j \in \mathbb{Z}. \tag{3.15}$$

For some constants $C > 0$ and $\beta > 1$, determined by a fixed annulus of analytic continuation,

$$|\widehat{q}_\ell| \leq C\beta^{-|\ell|}, \qquad \ell \in \mathbb{Z}.$$

Proof. Under the unitary pullback, H is multiplication by the positive function q, for which $d^2 \leq q \leq L^2$. The standard norm formula for multiplication operators gives (3.14). Fourier orthogonality gives

$$\langle HE_j, E_k\rangle_{\mathcal{H}_\Omega} = \frac{1}{2\pi}\int_0^{2\pi} q(e^{i\theta}) e^{-i(k-j)\theta}\, d\theta = \widehat{q}_{k-j}.$$

By Lemma 2.1, q is holomorphic on an annulus about the unit circle. Choose $\beta > 1$ so that $\beta^{-1} \leq |w| \leq \beta$ lies inside this annulus. Cauchy's Laurent coefficient formula on the two boundary circles gives

$$|\widehat{q}_\ell| \leq C\beta^{-|\ell|}, \qquad C \stackrel{\text{def}}{=} \max_{|w|\in\{\beta^{-1},\beta\}} |q(w)| > 0.$$

□

Remark 3.9. The scalar $\widehat{q}_0$ is the mean squared boundary radius with respect to harmonic measure, and $H - \widehat{q}_0 I$ measures its deviation from that mean. Although M_H has exponentially decaying off-diagonal entries, every column has squared norm $\sum_\ell |\widehat{q}_\ell|^2 > 0$. Hence $H \notin \mathcal{S}_2$.

For the r-th boundary correction,

$$\overline{\eta}^{\,r} T_\Omega^r f_b = |\eta|^{2r} \eta^{-r} T_\Omega^r f_b.$$

This separates the geometric multiplier from the normalized primitive. We now consider the following operators.

Proposition 3.10. *For $r \geq 1$, define*

$$\mathcal{R}_r^+ : H^2_{\mathrm{hol}}(\Omega) \longrightarrow H^2_{\mathrm{hol}}(\Omega), \quad (\mathcal{R}_r^+ f)(\eta) \stackrel{\mathrm{def}}{=} \eta^{-r} T_\Omega^r f(\eta), \tag{3.16}$$

where the value at zero is defined by removable continuation. For $b \in \mathcal{H}_\Omega$, with splitting $h_b = f_b + \overline{g_b}$, define

$$\begin{aligned} &\mathcal{R}_r : \mathcal{H}_\Omega \longrightarrow \mathcal{H}_\Omega, \\ &\mathcal{R}_r(b) \stackrel{\mathrm{def}}{=} \left(\mathcal{R}_r^+(f_b) + \overline{\mathcal{R}_r^+(g_b)} \right)\Big|_{\partial\Omega}, \qquad \mathcal{R}_0 \stackrel{\mathrm{def}}{=} I_{\mathcal{H}_\Omega}. \end{aligned} \tag{3.17}$$

Then $\mathcal{R}_r^+$ and $\mathcal{R}_r$ are complex-linear Hilbert–Schmidt operators and

$$\|\mathcal{R}_r^+\|_{\mathcal{S}_2} \leq \frac{\pi}{\sqrt{6}} \left(\frac{\|\phi'\|_\infty}{d} \right)^r, \qquad \|\mathcal{R}_r\|_{\mathcal{S}_2}^2 \leq \frac{1}{(r!)^2} + \frac{\pi^2}{3} \left(\frac{\|\phi'\|_\infty}{d} \right)^{2r}. \tag{3.18}$$

With

$$C_0 \stackrel{\mathrm{def}}{=} \sqrt{1 + \pi^2/3}, \qquad C_\Omega \stackrel{\mathrm{def}}{=} \max\{1, \|\phi'\|_\infty / d\},$$

one has $\|\mathcal{R}_r\|_{\mathcal{S}_2} \leq C_0 C_\Omega^r$. Moreover,

$$\mathcal{R}_r(\eta^k) = \frac{k!}{(k+r)!} \eta^k, \qquad \mathcal{R}_r(S^k) = \frac{k!}{(k+r)!} S^k, \qquad k \geq 0, \tag{3.19}$$

where the constant mode is included only once.

Proof. The zero of $T_\Omega^r f$ at the origin makes the quotient in (3.16) holomorphic. If $F_r = (T_\Omega^r f) \circ \phi$, then

$$(\mathcal{R}_r^+ f) \circ \phi = \left(\frac{w}{\phi(w)} \right)^r \frac{F_r(w)}{w^r}.$$

Division by w^r is an isometry from $w^r H^2$ onto H^2, and the maximum modulus principle gives $\|(w/\phi)^r\|_\infty = d^{-r}$. Hence

$$\|\mathcal{R}_r^+\|_{\mathcal{S}_2} \leq d^{-r} \|T_\Omega^r\|_{\mathcal{S}_2},$$

which proves the first bound. In the orthonormal basis $(E_j)_{j \in \mathbb{Z}}$, the constant column has norm $1/r!$, and the positive and negative frequency blocks have equal Hilbert–Schmidt norms. Therefore

$$\|\mathcal{R}_r\|_{\mathcal{S}_2}^2 = \frac{1}{(r!)^2} + 2 \sum_{k \geq 1} \|\mathcal{R}_r^+(\psi^k)\|_{H^2_{\mathrm{hol}}}^2 \leq \frac{1}{(r!)^2} + 2 \|\mathcal{R}_r^+\|_{\mathcal{S}_2}^2,$$

and (3.18) follows. Formula (3.19) is immediate from (3.5); the reflected formula uses $S^k = \overline{\eta}^{\,k}$ on the boundary. □

By construction,

$$H^r \mathcal{R}_r(b) = \overline{\eta}^{\,r} T_\Omega^r f_b(\eta) + \eta^r \overline{T_\Omega^r g_b(\eta)} \quad \text{on } \partial\Omega. \tag{3.20}$$

Indeed, $|\eta|^{2r} \eta^{-r} = \overline{\eta}^{\,r}$ and $|\eta|^{2r} \overline{\eta}^{-r} = \eta^r$ on the boundary.

3.4 Centered expansion of the Wronskian

Since $\mathfrak{B}_k^{(n)} = \mathfrak{B}_{k+n}$, Taylor expansion at the mean squared radius gives

$$\mathfrak{B}_k\left(-\frac{\lambda H}{4}\right) = \sum_{n=0}^{\infty} \frac{(-\lambda/4)^n}{n!}(H - \widehat{q}_0 I)^n \mathfrak{B}_{k+n}\left(-\frac{\lambda \widehat{q}_0}{4}\right). \tag{3.21}$$

The functions $\mathfrak{B}_k$ are entire, so no smallness condition on $H - \widehat{q}_0 I$ is needed.

Define the normalized boundary monomials

$$\beta_p(\eta) \stackrel{\text{def}}{=} \begin{cases} \eta^p/p!, & p \geq 0, \\ S(\eta)^{-p}/(-p)!, & p < 0, \end{cases}$$

and the Bessel coefficient operators

$$D_n(\lambda) \stackrel{\text{def}}{=} \sum_{m=0}^{\infty} \frac{(-\lambda \widehat{q}_0/4)^m}{m!} \mathcal{R}_{n+m}, \qquad n \geq 0. \tag{3.22}$$

Proposition 3.11. *For every $n \in \mathbb{N}_0$ and $\lambda \in \mathbb{C}$, the series* (3.22) *defines a bounded operator $D_n(\lambda) \in \mathcal{B}(\mathcal{H}_\Omega)$. For $n \geq 1$, D_n is entire with values in $\mathcal{S}_2(\mathcal{H}_\Omega)$, while $D_0 - I$ has the same property. More precisely,*

$$\|D_n(\lambda)\|_{\mathcal{S}_2} \leq C_0 C_\Omega^n \exp\left(\frac{|\lambda| \widehat{q}_0 C_\Omega}{4}\right), \qquad n \geq 1, \tag{3.23}$$

$$\|D_0(\lambda) - I\|_{\mathcal{S}_2} \leq C_0 \left[\exp\left(\frac{|\lambda| \widehat{q}_0 C_\Omega}{4}\right) - 1\right]. \tag{3.24}$$

For every $p \in \mathbb{Z}$,

$$D_n(\lambda)\beta_p = |p|! \mathfrak{B}_{|p|+n}\left(-\frac{\lambda \widehat{q}_0}{4}\right)\beta_p. \tag{3.25}$$

Proof. For $n \geq 1$, Proposition 3.10 gives

$$\sum_{m \geq 0} \left\| \frac{(-\lambda \widehat{q}_0/4)^m}{m!} \mathcal{R}_{n+m} \right\|_{\mathcal{S}_2} \leq C_0 C_\Omega^n \sum_{m \geq 0} \frac{(|\lambda| \widehat{q}_0 C_\Omega/4)^m}{m!},$$

which is (3.23). For $n = 0$, the $m = 0$ term is $\mathcal{R}_0 = I$; summing the remaining terms from $m = 1$ gives (3.24). The same majorants are uniform for λ in compact sets, so the series define entire functions in the stated operator ideals.

If $k = |p|$, then (3.19) yields

$$D_n(\lambda)\beta_p = \sum_{m \geq 0} \frac{(-\lambda \widehat{q}_0/4)^m}{m!} \frac{k!}{(k+n+m)!} \beta_p,$$

which is (3.25). □

Proposition 3.12. *For every $\lambda \in \mathbb{C}$, the boundary Wronskian satisfies*

$$W(\lambda) = \sum_{n=0}^{\infty} \frac{(-\lambda/4)^n}{n!}(H - \widehat{q}_0 I)^n D_n(\lambda), \tag{3.26}$$

with locally uniform operator-norm convergence. Moreover,

$$K(\lambda) \stackrel{\text{def}}{=} W(\lambda) - I_{\mathcal{H}_\Omega} \in \mathcal{S}_2(\mathcal{H}_\Omega), \tag{3.27}$$

and $K : \mathbb{C} \to \mathcal{S}_2(\mathcal{H}_\Omega)$ *is entire, with* $K(0) = 0$ *and*

$$\|K(\lambda)\|_{\mathcal{S}_2} \le C_0 \left[\exp\left(\frac{|\lambda| L^2 C_\Omega}{4}\right) - 1\right]. \tag{3.28}$$

Proof. Separate $D_0 = I + (D_0 - I)$. For $n \ge 1$, the ideal property and (3.23) give

$$\left\| \frac{(-\lambda/4)^n}{n!} (H - \widehat{q}_0 I)^n D_n(\lambda) \right\|_{\mathcal{S}_2} \le C_0 e^{|\lambda| \widehat{q}_0 C_\Omega/4} \frac{(|\lambda| C_\Omega \|H - \widehat{q}_0 I\|/4)^n}{n!}.$$

This majorant is locally summable. Together with (3.24), it proves that the centered series is an identity plus an entire Hilbert–Schmidt family.

To identify the sum, substitute (3.22) and first remove the term $(n, m) = (0, 0)$, which is the identity. The remaining double series is absolutely convergent in $\mathcal{S}_2$, because the sum of its norm majorants is bounded by

$$C_0 \left[\exp\left(\frac{|\lambda| C_\Omega}{4} (\|H - \widehat{q}_0 I\| + \widehat{q}_0)\right) - 1\right].$$

It may therefore be grouped by $r = n + m$. The coefficient of $\mathcal{R}_r$ is

$$\begin{aligned} &\left(-\frac{\lambda}{4}\right)^r \sum_{n=0}^{r} \frac{(H - \widehat{q}_0 I)^n \widehat{q}_0^{\,r-n}}{n!(r-n)!} \\ &\qquad = \frac{(-\lambda/4)^r}{r!} \sum_{n=0}^{r} \binom{r}{n} (H - \widehat{q}_0 I)^n (\widehat{q}_0 I)^{r-n} = \frac{(-\lambda/4)^r}{r!} H^r. \end{aligned}$$

Here the two multiplier factors commute, and $\mathcal{R}_r$ remains on their right. Hence (3.20) gives

$$\sum_{n \ge 0} \frac{(-\lambda/4)^n}{n!} (H - \widehat{q}_0 I)^n D_n(\lambda) b = \sum_{r \ge 0} \frac{(-\lambda/4)^r}{r!} H^r \mathcal{R}_r(b) = W(\lambda) b.$$

Finally,

$$\begin{aligned} \|K(\lambda)\|_{\mathcal{S}_2} &\le \sum_{r \ge 1} \frac{(|\lambda|/4)^r}{r!} \|H\|^r \|\mathcal{R}_r\|_{\mathcal{S}_2} \\ &\le C_0 \sum_{r \ge 1} \frac{(|\lambda| L^2 C_\Omega/4)^r}{r!}, \end{aligned}$$

which proves (3.28) and $K(0) = 0$. □

Proposition 3.13. *For* $p \in \mathbb{Z}$ *and* $\lambda \in \mathbb{C}$*, the Wronskian sends the normalized harmonic monomial* β_p *to the Bessel boundary trace*

$$W(\lambda)\beta_p = \Phi_p(\eta; \lambda), \qquad \Phi_p(\eta; \lambda) = \begin{cases} \eta^p \mathfrak{B}_p(-\lambda \eta S(\eta)/4), & p \ge 0, \\ S(\eta)^{-p} \mathfrak{B}_{-p}(-\lambda \eta S(\eta)/4), & p < 0. \end{cases} \tag{3.29}$$

Proof. Let $k = |p|$. Equations (3.25) and (3.26), followed by the Taylor identity (3.21), give

$$W(\lambda)\beta_p = k! \beta_p \, \mathfrak{B}_k(-\lambda H/4).$$

Now $k!\beta_p = \eta^p$ for $p \ge 0$ and $k!\beta_p = S^{-p}$ for $p < 0$, while $H = \eta S$. □

3.5 Sobolev regularity and the spectral criterion

Since $W(\lambda) = I + K(\lambda)$ with $K(\lambda) \in \mathcal{S}_2(\mathcal{H}_\Omega)$, the regularized determinant $\det{}_2 W(\lambda)$ is well defined. To interpret its zeros spectrally, a kernel datum initially known only to lie in $\mathcal{H}_\Omega$ must acquire enough regularity to produce an H^1_0 solution.

Proposition 3.14. *For every $\lambda \in \mathbb{C}$, the maps*

$$K(\lambda) : \mathcal{H}_\Omega \longrightarrow H^1(\partial\Omega)$$

and

$$\mathcal{H}_\Omega \longrightarrow H^1(\Omega), \qquad b \longmapsto u_b(\cdot;\lambda) - h_b,$$

are bounded, and their norms are locally uniformly bounded with respect to λ. If $b \in H^{1/2}(\partial\Omega)$, then $u_b \in H^1(\Omega)$ and

$$\gamma u_b = W(\lambda) b.$$

Proof. Let $v_{r,b}$ denote the r-th bracket in (3.7). For $G_r = T^r_\Omega f$ and $F_r = G_r \circ \phi$, the primitive estimates and (2.6) imply

$$\|G_r\|_{H^1(\Omega)} \le C_1 \|\phi'\|^r_\infty \|f\|_{H^2_{\mathrm{hol}}}$$

for some $C_1 = C_1(\phi) > 0$. Since

$$|\overline{\eta}^r| \le L^r, \qquad |\nabla \overline{\eta}^r| \le \sqrt{2} r L^{r-1},$$

the product rule gives

$$\|\overline{\eta}^r G_r\|_{H^1(\Omega)} \le C_1 (L^r + \sqrt{2} r L^{r-1}) \|\phi'\|^r_\infty \|f\|_{H^2_{\mathrm{hol}}}.$$

On the boundary, differentiation of $\overline{\phi}^r F_r$ yields

$$\|\overline{\phi}^r F_r\|_{H^1(\partial\mathbb{D})} \le (2L^r + r L^{r-1} \|\phi'\|_\infty) \|\phi'\|^r_\infty \|f\|_{H^2_{\mathrm{hol}}}.$$

The reflected terms satisfy the same estimates. Since $\|f_b\|_{H^2_{\mathrm{hol}}} + \|g_b\|_{H^2_{\mathrm{hol}}} \le \sqrt{2}\|b\|_{\mathcal{H}_\Omega}$, there exist constants $C_2 > 0$ and $A_\Omega \ge 1$, depending only on the fixed domain and conformal map, such that

$$\|v_{r,b}\|_{H^1(\Omega)} + \|v_{r,b}|_{\partial\Omega}\|_{H^1(\partial\Omega)} \le C_2 (1 + r) A^r_\Omega \|b\|_{\mathcal{H}_\Omega}. \tag{3.30}$$

With $x = |\lambda| A_\Omega / 4$, the weighted sum satisfies

$$\sum_{r \ge 1} \frac{(|\lambda|/4)^r}{r!} (1 + r) A^r_\Omega = (e^x - 1) + x e^x.$$

Thus the correction series converges in both asserted Sobolev norms, locally uniformly in λ. Its boundary sum is $K(\lambda)b$, since the zeroth-order term is b. If $b \in H^{1/2}(\partial\Omega)$, then $h_b \in H^1(\Omega)$ by Lemma 3.1. Continuity of the trace map and uniqueness of the radial L^2 limit then give $\gamma u_b = W(\lambda) b$. □

Theorem 3.15. *For $\lambda > 0$, let*

$$E^D_\lambda \overset{\text{def}}{=} \{u \in H^1_0(\Omega) : -\Delta u = \lambda u\}.$$

The assignment $b \mapsto u_b$ is a complex-linear bijection from $\ker W(\lambda)$ onto E^D_λ. Consequently,

$$\lambda \text{ is a Dirichlet eigenvalue of } \Omega \quad \Longleftrightarrow \quad \ker W(\lambda) \neq \{0\}, \tag{3.31}$$

and

$$\dim \ker W(\lambda) = \dim E^D_\lambda < \infty. \tag{3.32}$$

Since the Dirichlet Laplacian is self-adjoint, $\dim E^D_\lambda$ is the spectral multiplicity of λ.

Proof. First let $b \in \ker W(\lambda)$. Since $W = I + K$,

$$b = -K(\lambda)b.$$

Proposition 3.14 places the right-hand side in $H^1(\partial\Omega)$, and hence $b \in H^{1/2}(\partial\Omega)$. The same proposition gives $u_b \in H^1(\Omega)$ and

$$\gamma u_b = W(\lambda)b = 0.$$

Thus $u_b \in H^1_0(\Omega)$. The interior equation holds classically and hence weakly, while Proposition 3.3 shows that $b \neq 0$ implies $u_b \neq 0$.

Conversely, let $u \in E^D_\lambda$. By Theorem 3.6, there is a unique $b \in H^{1/2}(\partial\Omega)$ such that $u = u_b$. The trace identity then gives

$$W(\lambda)b = \gamma u = 0.$$

If $u \neq 0$, injectivity of the construction forces $b \neq 0$. These two assignments are linear and inverse to one another, proving the bijection and the dimension identity. □

Theorem 3.16. *The function $\lambda \mapsto \det{}_2 W(\lambda)$ is entire and equals one at zero. For every $\lambda \in \mathbb{C}$,*

$$\det{}_2 W(\lambda) = 0 \quad \Longleftrightarrow \quad W(\lambda) \text{ is non-invertible} \quad \Longleftrightarrow \quad \ker W(\lambda) \neq \{0\}.$$

Consequently, for $\lambda > 0$,

$$\det{}_2 W(\lambda) = 0 \quad \Longleftrightarrow \quad \lambda \text{ is a Dirichlet eigenvalue of } \Omega. \tag{3.33}$$

Proof. Proposition 3.12 gives an entire map $K : \mathbb{C} \to \mathcal{S}_2(\mathcal{H}_\Omega)$, with $W = I + K$ and $K(0) = 0$. Lemma 2.9 therefore gives an entire scalar function, its value one at zero, and the two operator equivalences. For positive real λ, Theorem 3.15 identifies a nonzero kernel with a nonzero Dirichlet eigenfunction. □

Corollary 3.17. *Every zero of $\det{}_2 W$ is positive and real.*

Proof. If $\det{}_2 W(\lambda) = 0$, Lemma 2.9 supplies a nonzero $b \in \ker W(\lambda)$. The smoothing argument above, which also holds for complex λ, gives a nonzero $u_b \in H^1_0(\Omega)$ satisfying $-\Delta u_b = \lambda u_b$. Testing against u_b yields

$$\int_\Omega |\nabla u_b|^2 dA = \lambda \int_\Omega |u_b|^2 dA.$$

The denominator is positive. The numerator is also positive, since a function in $H^1_0(\Omega)$ with zero gradient is zero. Hence λ is positive and real. □

Remark 3.18. Writing $m_D(\lambda)$ for the spectral multiplicity of a Dirichlet eigenvalue, the kernel correspondence proves $m_D(\lambda) = \dim\ker W(\lambda)$. It does not show that this dimension equals the vanishing order of $\det{}_2 W$ at λ. Establishing such an equality requires a local analysis of the analytic operator family.

4 Wronskian Matrices from Riemann Mapping Coefficients

We now express the operator $W(\lambda)$ in terms of the coefficients of the normalized Riemann map. The infinite-dimensional formulas are established first; finite cutoffs are introduced only after each matrix entry has been identified.

4.1 Conformal coordinates and infinite matrices

Recall that $C_\phi b = b \circ \phi$ is unitary from $\mathcal{H}_\Omega$ onto $L^2(\partial\mathbb{D})$, and that $E_j = C_\phi^{-1} e_j$, where $e_j(e^{i\theta}) = e^{ij\theta}$. If $B \in \mathcal{B}(\mathcal{H}_\Omega)$, we write

$$[B]_{\widetilde{B};j,k} \stackrel{\text{def}}{=} \langle BE_k, E_j\rangle_{\mathcal{H}_\Omega} = \frac{1}{2\pi}\int_0^{2\pi} (BE_k)(\phi(e^{i\theta}))e^{-ij\theta}\,d\theta. \tag{4.1}$$

Thus $[B]_{\widetilde{B}}$ is the Fourier matrix of $C_\phi B C_\phi^{-1}$. Unitary equivalence gives

$$\|[B]_{\widetilde{B}}\| = \|B\|, \qquad \|[B]_{\widetilde{B}}\|_{\mathcal{S}_2} = \|B\|_{\mathcal{S}_2} \quad (B \in \mathcal{S}_2),$$

and $[B_1B_2]_{\widetilde{B}} = [B_1]_{\widetilde{B}}[B_2]_{\widetilde{B}}$. We apply this notation to $W(\lambda)$.

Write

$$\phi(w) = \sum_{j\ge1} a_j w^j, \qquad a_1 > 0, \tag{4.2}$$

and recall that $q(e^{i\theta}) = |\phi(e^{i\theta})|^2$. The function q is the conformal-coordinate symbol of the boundary multiplication operator H, so its Fourier coefficients give the matrix of H.

Proposition 4.1. *The Fourier coefficients of q are*

$$\widehat{q}_\ell = \sum_{m\ge1} a_{m+\ell}\overline{a_m} \quad (\ell \ge 0), \qquad \widehat{q}_{-\ell} = \overline{\widehat{q}_\ell}, \qquad \widehat{q}_0 = \sum_{m\ge1}|a_m|^2. \tag{4.3}$$

The sums are absolutely convergent, and the multiplication operator $Hb = |\eta|^2 b$ has the Toeplitz matrix

$$M_H = [H]_{\widetilde{B}}, \qquad (M_H)_{j,k} = \widehat{q}_{j-k}. \tag{4.4}$$

Proof. Because $\partial\Omega$ is real analytic, ϕ is holomorphic in $|w| < R_\phi$ for some $R_\phi > 1$. For $1 < \tau < R_\phi$, Cauchy's estimate gives

$$|a_j| \le C_\tau \tau^{-j}, \qquad C_\tau \stackrel{\text{def}}{=} \max_{|w|=\tau}|\phi(w)|.$$

Hence $(a_j) \in \ell^1$, and the product

$$q(e^{i\theta}) = \sum_{j,m\ge1} a_j\overline{a_m}e^{i(j-m)\theta}$$

is absolutely convergent. Collecting the terms with $j - m = \ell$ gives (4.3); in particular,

$$\sum_{m\ge1}|a_{m+\ell}a_m| \le \frac{C_\tau^2}{\tau^2-1}\tau^{-\ell}.$$

The Toeplitz formula follows by taking Fourier coefficients of qe_k. Since $q \in L^\infty(\partial\mathbb{D})$, M_H is bounded on $\ell^2(\mathbb{Z})$. □

We next convert the primitive operator into conformal power-series coordinates. For $F = f \circ \phi$, the change of variables $\zeta = \phi(\xi)$ gives

$$(T_\Omega f)(\phi(w)) = \int_0^w f(\phi(\xi))\phi'(\xi)\,d\xi.$$

We therefore define

$$(\mathcal{T}_\phi F)(w) \stackrel{\text{def}}{=} \int_0^w F(\xi)\phi'(\xi)\,d\xi. \tag{4.5}$$

Thus $\mathcal{T}_\phi$ is the conformal-coordinate representation of the primitive operator T_Ω.

Proposition 4.2. *If $F \in H^2(\mathbb{D})$, with*

$$F(w) = \sum_{n\geq 0} x_n w^n, \qquad \phi'(w) = \sum_{j\geq 1} j a_j w^{j-1},$$

then

$$\mathcal{T}_\phi F(w) = \sum_{n\geq 0}\sum_{j\geq 1} \frac{j a_j}{n+j} x_n w^{n+j}.$$

Consequently, relative to $(1, w, w^2, \ldots)$,

$$[\mathcal{T}_\phi]_{m,n} = \begin{cases} \dfrac{m-n}{m} a_{m-n}, & m > n, \\ 0, & m \leq n. \end{cases} \tag{4.6}$$

Moreover, $\mathcal{T}_\phi$ is bounded on $H^2(\mathbb{D})$ and $\|\mathcal{T}_\phi\| \leq \|\phi'\|_\infty$.

Proof. Termwise multiplication and integration give

$$\begin{aligned} \mathcal{T}_\phi F(w) &= \int_0^w \left(\sum_{n\geq 0} x_n \xi^n\right)\left(\sum_{j\geq 1} j a_j \xi^{j-1}\right) d\xi \\ &= \sum_{n\geq 0}\sum_{j\geq 1} \frac{j a_j}{n+j} x_n w^{n+j}. \end{aligned}$$

Putting $m = n + j$, the coefficient of w^m is

$$\sum_{n=0}^{m-1} \frac{m-n}{m} a_{m-n} x_n,$$

which proves (4.6). Finally, if $F\phi' = \sum_{n\geq 0} g_n w^n$, then

$$\|\mathcal{T}_\phi F\|_{H^2}^2 = \sum_{n\geq 0} \frac{|g_n|^2}{(n+1)^2} \leq \|F\phi'\|_{H^2}^2 \leq \|\phi'\|_\infty^2 \|F\|_{H^2}^2.$$

□

The operators $\mathcal{R}_r$ isolate the normalized primitive part of the boundary corrections constructed in Section 3. We now compute their matrices from the coefficients of ϕ. For $r, k \in \mathbb{N}_0$, set

$$F_{0,k}(w) \stackrel{\text{def}}{=} w^k, \qquad F_{r+1,k} \stackrel{\text{def}}{=} \mathcal{T}_\phi F_{r,k}, \qquad F_{r,k}(w) = \sum_{\ell\geq 0} t_{\ell,k}^{(r)} w^\ell. \tag{4.7}$$

Then $F_{r,k} = \mathcal{T}_\phi^r(w^k)$ and

$$(\mathcal{R}_r^+ \psi^k)(\phi(w)) = \frac{F_{r,k}(w)}{\phi(w)^r}.$$

Every application of $\mathcal{T}_\phi$ adds a zero at $w = 0$. Since ϕ has a simple zero there and no other zero on a neighborhood of $\overline{\mathbb{D}}$, this quotient is holomorphic. Define

$$\left(\frac{w}{\phi(w)}\right)^r = \sum_{\ell\geq 0} c_\ell^{(r)} w^\ell, \qquad \frac{F_{r,k}(w)}{\phi(w)^r} = \sum_{\ell\geq 0} s_{\ell,k}^{(r)} w^\ell.$$

Here $t^{(r)}$ records the iterated primitive, $c^{(r)}$ records the reciprocal factor $(w/\phi(w))^r$, and their convolution $s^{(r)}$ gives the matrix entries of $\mathcal{R}_r$.

Proposition 4.3. *For $r, k, \ell \in \mathbb{N}_0$, the coefficients needed to construct $\mathcal{R}_r$ satisfy the finite recursions*

$$t_{\ell,k}^{(0)} = \delta_{\ell k}, \qquad t_{0,k}^{(r+1)} = 0, \qquad t_{\ell,k}^{(r+1)} = \frac{1}{\ell}\sum_{j=1}^{\ell} j a_j t_{\ell-j,k}^{(r)} \quad (\ell \geq 1), \tag{4.8}$$

$$s_{\ell,k}^{(r)} = \sum_{j=0}^{\ell} c_j^{(r)} t_{\ell-j+r,k}^{(r)}. \tag{4.9}$$

The reciprocal coefficients are obtained from

$$\begin{aligned} c_0^{(1)} &= a_1^{-1}, \qquad & c_\ell^{(1)} &= -a_1^{-1}\sum_{j=1}^{\ell} a_{j+1} c_{\ell-j}^{(1)} \quad (\ell \geq 1), \\ c_\ell^{(r+1)} &= \sum_{j=0}^{\ell} c_j^{(r)} c_{\ell-j}^{(1)}, & c_\ell^{(0)} &= \delta_{\ell 0}. \end{aligned} \tag{4.10}$$

For $r \geq 1$, the full boundary matrix is

$$[\mathcal{R}_r]_{\widetilde{B};j,k} = \begin{cases} 1/r!, & j = k = 0, \\ s_{j,k}^{(r)}, & j, k \geq 1, \\ s_{|j|,|k|}^{(r)}, & j, k \leq -1, \\ 0, & \textit{otherwise}. \end{cases} \tag{4.11}$$

For $r = 0$, this matrix is the identity. In addition,

$$s_{\ell,k}^{(r)} = 0 \quad (\ell < k), \qquad s_{k,k}^{(r)} = \frac{k!}{(k+r)!}, \qquad s_{\ell,0}^{(r)} = \frac{\delta_{\ell 0}}{r!}.$$

Proof. The coefficient of $w^{\ell-1}$ in $\phi' F_{r,k}$ is $\sum_{j=1}^{\ell} j a_j t_{\ell-j,k}^{(r)}$. Integration divides it by ℓ, proving (4.8). Since $F_{r,k}$ vanishes to order $k + r$,

$$\frac{F_{r,k}(w)}{\phi(w)^r} = \left(\sum_{j\geq 0} c_j^{(r)} w^j\right)\left(\sum_{\ell \geq 0} t_{\ell+r,k}^{(r)} w^\ell\right),$$

and coefficient comparison gives (4.9). Likewise,

$$\left(\sum_{j\geq 0} a_{j+1} w^j\right)\left(\sum_{\ell \geq 0} c_\ell^{(1)} w^\ell\right) = 1$$

gives (4.10).

The leading term follows from $\psi(\eta) = a_1^{-1}\eta + O(\eta^2)$:

$$T_\Omega^r(\psi^k)(\eta) = a_1^{-k}\frac{k!}{(k+r)!}\eta^{k+r} + O(\eta^{k+r+1}).$$

After division by η^r and composition with ϕ, this yields the vanishing and diagonal identities. The case $k = 0$ follows from $T_\Omega^r 1 = \eta^r/r!$. Finally, $E_{-k} = \overline{E_k}$ on the boundary, so reflection gives the negative block in (4.11); the holomorphic and antiholomorphic parts do not mix. □

The operators $D_n(\lambda)$ collect the normalized primitive operators in the centered expansion. Substitution of their matrices now gives the matrix of the boundary operator $W(\lambda)$.

Proposition 4.4. *For $n \in \mathbb{N}_0$ and $\lambda \in \mathbb{C}$, the conformal-basis matrices satisfy*

$$[D_n(\lambda)]_{\widetilde{B}} = \sum_{m\geq 0} \frac{(-\lambda\widehat{q}_0/4)^m}{m!}[\mathcal{R}_{n+m}]_{\widetilde{B}}, \tag{4.12}$$

$$[W(\lambda)]_{\widetilde{B}} = \sum_{n\geq 0} \frac{(-\lambda/4)^n}{n!}(M_H - \widehat{q}_0 I)^n [D_n(\lambda)]_{\widetilde{B}} \tag{4.13}$$

$$= \sum_{r\geq 0} \frac{(-\lambda/4)^r}{r!} M_H^r [\mathcal{R}_r]_{\widetilde{B}}. \tag{4.14}$$

These series have the operator-ideal convergence established in Section 3. In particular,

$$[D_n(\lambda)]_{\widetilde{B};k,k} = |k|!\mathfrak{B}_{|k|+n}\left(-\frac{\lambda\widehat{q}_0}{4}\right).$$

Proof. The coefficient map is unitary and multiplicative. Applying it to the operator formulas for $D_n(\lambda)$ and $W(\lambda)$ gives (4.12)–(4.14), with the same norm convergence. The diagonal formula follows from (4.11):

$$\begin{aligned}[D_n(\lambda)]_{\widetilde{B};k,k} &= \sum_{m\geq 0} \frac{(-\lambda\widehat{q}_0/4)^m}{m!}\frac{|k|!}{(|k|+n+m)!} \\ &= |k|!\mathfrak{B}_{|k|+n}\left(-\frac{\lambda\widehat{q}_0}{4}\right).\end{aligned}$$

Thus the data $(a_j)_{j\geq 1}$ determine q, M_H, $\mathcal{T}_\phi$, every $\mathcal{R}_r$, and finally every entry of $[W(\lambda)]_{\widetilde{B}}$. Since C_ϕ is unitary, this matrix determines $W(\lambda)$ itself and has the same kernel and regularized determinant. □

4.2 Finite construction and fixed-parameter consistency

Let $A \in \mathbb{N}$ truncate the Riemann-map coefficients, $J \in \mathbb{N}_0$ the inner series in D_n, $M \in \mathbb{N}_0$ the centered Wronskian series, and $N \in \mathbb{N}_0$ the reported Fourier modes $I_N = \{-N, \dots, N\}$. Set

$$a_j^{(A)} \stackrel{\text{def}}{=} \begin{cases} a_j, & 1 \leq j \leq A, \\ 0, & j > A, \end{cases} \qquad \phi_A(w) = \sum_{j=1}^{A} a_j w^j. \tag{4.15}$$

Let $\mathcal{T}_{\phi_A}$ denote the operator in (4.5) with ϕ replaced by ϕ_A. Then $q_A = |\phi_A|^2$ has coefficients

$$\widehat{q}_\ell^{(A)} = \begin{cases} \displaystyle\sum_{m=1}^{A-\ell} a_{m+\ell}\overline{a_m}, & 0 \leq \ell \leq A-1, \\ 0, & \ell \geq A, \end{cases} \qquad \widehat{q}_{-\ell}^{(A)} = \overline{\widehat{q}_\ell^{(A)}}. \tag{4.16}$$

The Cauchy estimate used above implies $\phi_A \to \phi$ and $\phi_A' \to \phi'$ uniformly on $\overline{\mathbb{D}}$, and consequently $q_A \to q$ uniformly. To control the reciprocal factors used below, set

$$\chi(w) \stackrel{\text{def}}{=} \begin{cases} \phi(w)/w, & w \neq 0, \\ \phi'(0), & w = 0, \end{cases} \qquad \chi_A(w) \stackrel{\text{def}}{=} \begin{cases} \phi_A(w)/w, & w \neq 0, \\ a_1, & w = 0. \end{cases}$$

The function χ is continuous and nonzero on $\overline{\mathbb{D}}$, since ϕ has no zero there except at the origin. Hence

$$\delta_\phi \stackrel{\text{def}}{=} \min_{|w|\le 1} |\chi(w)| > 0.$$

The uniform convergence $\chi_A \to \chi$ gives an integer $A_0 \ge 1$ such that, for every $A \ge A_0$,

$$\inf_{|w|\le 1} |\chi_A(w)| \ge \frac{\delta_\phi}{2}.$$

Moreover,

$$\|\chi_A^{-1} - \chi^{-1}\|_\infty \le \frac{2}{\delta_\phi^2}\|\chi_A - \chi\|_\infty \longrightarrow 0.$$

Thus, for each fixed $r \in \mathbb{N}$, $(w/\phi_A(w))^r \to (w/\phi(w))^r$ in $H^\infty(\mathbb{D})$.

For $A \ge A_0$, replace a_j by $a_j^{(A)}$ in (4.8), (4.9), and (4.10). Denote the resulting block matrix by $\mathbf{R}_r^{(A)} = (R_{r;j,k}^{(A)})$, using (4.11), and put $\mathbf{R}_0^{(A)} = I$. We then define

$$[D_{n,J}^{(A)}(\lambda)]_{\ell,k} \stackrel{\text{def}}{=} \sum_{m=0}^{J} \frac{(-\lambda \widehat{q}_0^{(A)}/4)^m}{m!} R_{n+m;\ell,k}^{(A)}. \tag{4.17}$$

For the centered multiplier, set

$$\mu_s^{(n,A)} \stackrel{\text{def}}{=} \widehat{(q_A - \widehat{q}_0^{(A)})^n}_s, \qquad \mu_s^{(0,A)} = \delta_{s0}.$$

Its Fourier support lies in $|s| \le n(A-1)$. Hence, for $j, k \in I_N$, every sum in

$$\boxed{[W_{N,M,J}^{(A)}(\lambda)]_{j,k} = \sum_{n=0}^{M} \frac{(-\lambda/4)^n}{n!} \sum_{\ell=j-n(A-1)}^{j+n(A-1)} \mu_{j-\ell}^{(n,A)} [D_{n,J}^{(A)}(\lambda)]_{\ell,k}} \tag{4.18}$$

is finite and defines a $(2N+1)\times(2N+1)$ matrix indexed by I_N.

The truncation of the two Taylor series is controlled by the factorial tails

$$\left\| D_n(\lambda) - \sum_{m=0}^{J} \frac{(-\lambda \widehat{q}_0/4)^m}{m!} \mathcal{R}_{n+m} \right\|_{\mathcal{S}_2} \le C_0 C_\Omega^n \sum_{m>J} \frac{(|\lambda| \widehat{q}_0 C_\Omega/4)^m}{m!}, \tag{4.19}$$

$$\left\| W(\lambda) - \sum_{n=0}^{M} \frac{(-\lambda/4)^n}{n!} (H - \widehat{q}_0 I)^n D_n(\lambda) \right\|_{\mathcal{S}_2} \le C_0 e^{|\lambda| \widehat{q}_0 C_\Omega/4} \sum_{n>M} \frac{(|\lambda| C_\Omega \|q - \widehat{q}_0\|_\infty/4)^n}{n!}, \tag{4.20}$$

where $C_0, C_\Omega > 0$ are the constants defined in Proposition 3.10.

For an $m \times m$ matrix B, we use

$$\det{}_2 B \stackrel{\text{def}}{=} \det B \exp\big(-\operatorname{tr}(B - I_m)\big). \tag{4.21}$$

The exponential factor never vanishes, so $\det_2 B$ and $\det B$ have the same zeros.

To separate the Taylor cutoffs from the geometric cutoff, define

$$[D_{n,J}(\lambda)]_{\widetilde{B};\ell,k} \stackrel{\text{def}}{=} \sum_{m=0}^{J} \frac{(-\lambda \widehat{q}_0/4)^m}{m!} [\mathcal{R}_{n+m}]_{\widetilde{B};\ell,k}$$

and, for $j, k \in I_N$,

$$[W^{(\infty)}_{N,M,J}(\lambda)]_{j,k} \stackrel{\text{def}}{=} \sum_{n=0}^{M} \frac{(-\lambda/4)^n}{n!} \sum_{\ell\in\mathbb{Z}} \widehat{(q-\widehat{q}_0)^n}_{\,j-\ell} [D_{n,J}(\lambda)]_{\widetilde{B};\ell,k}. \tag{4.22}$$

The inner sum is the (j,k)-entry of $(M_H - \widehat{q}_0 I)^n [D_{n,J}(\lambda)]_{\widetilde{B}}$ and converges absolutely by Cauchy–Schwarz.

Proposition 4.5. *For each fixed $\lambda \in \mathbb{C}$, the finite construction is consistent in the following sequential sense.*

1. *Let Π_N project onto $\operatorname{span}\{E_j : |j| \le N\}$, and let $W^{\mathrm{ex}}_N(\lambda)$ be the matrix of $I + \Pi_N K(\lambda)\Pi_N$ on $\Pi_N \mathcal{H}_\Omega$. Equivalently, it is the corresponding principal block of $[W(\lambda)]_{\widetilde{B}}$. Then*

$$\|K(\lambda) - \Pi_N K(\lambda)\Pi_N\|_{\mathcal{S}_2} \to 0, \qquad \det{}_2 W^{\mathrm{ex}}_N(\lambda) \to \det{}_2 W(\lambda).$$

2. *For fixed N, M, J, the exact-coefficient block in* (4.22) *satisfies*

$$\|W^{(A)}_{N,M,J}(\lambda) - W^{(\infty)}_{N,M,J}(\lambda)\|_{\mathrm{F}} \to 0 \qquad (A \to \infty,\ A \ge A_0), \tag{4.23}$$

   and the corresponding regularized determinants converge.

3. *For this λ, one can choose integer sequences $N_\nu, M_\nu, J_\nu, A_\nu \to \infty$ such that*

$$\det{}_2 W^{(A_\nu)}_{N_\nu,M_\nu,J_\nu}(\lambda) \longrightarrow \det{}_2 W(\lambda). \tag{4.24}$$

Here $\|B\|_{\mathrm{F}} \stackrel{\text{def}}{=} (\sum_{j,k} |B_{j,k}|^2)^{1/2}$ denotes the Frobenius norm.

Proof. We first truncate the Hilbert–Schmidt operator $K(\lambda) = W(\lambda) - I$. In the conformal orthonormal basis,

$$\|K - \Pi_N K \Pi_N\|^2_{\mathcal{S}_2} = \sum_{\substack{j,k\in\mathbb{Z} \\ |j|>N \text{ or } |k|>N}} |[K]_{\widetilde{B};j,k}|^2 \longrightarrow 0.$$

The last expression is the tail of the convergent series defining $\|K\|^2_{\mathcal{S}_2}$. Hence it tends to zero. By the $\mathcal{S}_2$-continuity of $\det_2(I + \cdot)$, we obtain the first determinant limit.

For fixed r, uniform convergence of ϕ'_A gives

$$\|\mathcal{T}_{\phi_A} - \mathcal{T}_\phi\|_{\mathcal{S}_2} \le \frac{\pi}{\sqrt{6}} \|\phi'_A - \phi'\|_\infty \longrightarrow 0.$$

The identity

$$\mathcal{T}^r_{\phi_A} - \mathcal{T}^r_\phi = \sum_{h=0}^{r-1} \mathcal{T}^h_{\phi_A} (\mathcal{T}_{\phi_A} - \mathcal{T}_\phi) \mathcal{T}^{r-1-h}_\phi$$

and the ideal estimate in Lemma 2.8 show that each power converges in $\mathcal{S}_2$. Combining this with the uniform convergence of the reciprocal factors, and using that division by w^r is an isometry on $w^r H^2$, we obtain

$$\|\mathbf{R}^{(A)}_r - [\mathcal{R}_r]_{\widetilde{B}}\|_{\mathcal{S}_2} \to 0 \qquad (r \ge 1).$$

Since $q_A \to q$ in L^∞, every term in the fixed finite sums $n \le M$, $m \le J$ converges. Restricting the output and input indices to I_N gives (4.23). The matrix dimension is now fixed, so Frobenius convergence implies entrywise convergence and continuity of the finite-dimensional determinant gives the stated determinant limit.

For the final claim, fix ν. We first choose N_ν so that the first truncation error is below $1/\nu$. We then choose M_ν and J_ν from (4.20) and (4.19), and finally choose A_ν from (4.23). The choices may be enlarged so that every parameter is at least ν. The triangle inequality and continuity of $\det_2$ then give (4.24). □

Proposition 4.5 asserts convergence at a fixed spectral parameter along a coordinated choice of cutoffs. It does not give convergence for arbitrary independent growth of N, M, J, A. It also does not imply convergence of the computed zeros or their multiplicities; that would require locally uniform convergence in λ on a complex neighborhood and a contour argument.

5 The Neumann Oval: A Numerical Experiment

Now we illustrate a numerical experiment based on the Neumann oval with $R = 1.8$. All numerical experiments in this section were implemented in R 4.6.0 using RStudio 2026.05.1+225.

5.1 Analytic data

For the Neumann-oval family, consider the Riemann map

$$\phi_R(w) = \frac{2Rw}{R^2 + w^2} = \sum_{m=0}^{\infty} 2(-1)^m R^{-(2m+1)} w^{2m+1}, \qquad R > 1. \tag{5.1}$$

Proposition 5.1. *For $R > 1$, ϕ_R is conformal on a neighborhood of $\overline{\mathbb{D}}$, and its image has a real-analytic Jordan boundary. Its geometric data are*

$$q(e^{i\theta}) = \frac{4R^2}{|R^2 + e^{2i\theta}|^2}, \qquad \widehat{q}_0 = \frac{4R^2}{R^4 - 1}, \tag{5.2}$$

$$\widehat{q}_{2\ell} = \widehat{q}_0(-R^{-2})^{|\ell|}, \qquad \widehat{q}_{2\ell+1} = 0 \quad (\ell \in \mathbb{Z}), \tag{5.3}$$

$$d = \frac{2R}{R^2 + 1}, \qquad L = \frac{2R}{R^2 - 1}. \tag{5.4}$$

For the primitive recursion,

$$\phi_R'(w) = \frac{2R(R^2 - w^2)}{(R^2 + w^2)^2}, \qquad \left(\frac{w}{\phi_R(w)}\right)^r = \frac{(R^2 + w^2)^r}{(2R)^r}. \tag{5.5}$$

Proof. Choose $1 < \rho < R$. There are no poles in $|w| < \rho$. Equality of the images of w_1, w_2 reduces to

$$(w_1 - w_2)(R^2 - w_1 w_2) = 0.$$

Since $|w_1 w_2| < \rho^2 < R^2$, this implies $w_1 = w_2$. Thus the map is conformal across the closed unit disk and has an analytic Jordan image boundary. On the unit circle,

$$q(w) = \frac{4}{R^2}\left(\sum_{m\ge 0}(-R^{-2})^m w^{2m}\right)\left(\sum_{j\ge 0}(-R^{-2})^j w^{-2j}\right).$$

For $\ell \geq 0$, the coefficient of $w^{2\ell}$ is

$$\frac{4}{R^2}\sum_{j\geq 0}(-R^{-2})^{j+\ell}(-R^{-2})^{j} = \frac{4}{R^2}\frac{(-R^{-2})^{\ell}}{1-R^{-4}} = \frac{4R^2}{R^4-1}(-R^{-2})^{\ell}.$$

Negative coefficients follow by conjugation, and odd coefficients vanish. This proves (5.2)–(5.3). Also,

$$|\phi_R(e^{i\theta})| = \frac{2R}{|R^2+e^{2i\theta}|}, \qquad R^2-1 \leq |R^2+e^{2i\theta}| \leq R^2+1.$$

Both extrema are attained. This proves (5.4). Finally,

$$\phi_R'(w) = \frac{2R(R^2+w^2)-4Rw^2}{(R^2+w^2)^2} = \frac{2R(R^2-w^2)}{(R^2+w^2)^2},$$

and $w/\phi_R(w) = (R^2+w^2)/(2R)$. These identities prove (5.5). □

For the exact rational map, the reciprocal coefficients in (4.9) are explicit:

$$c_{2j}^{(r)} = \binom{r}{j}\frac{R^{2(r-j)}}{(2R)^r} \quad (0 \leq j \leq r), \qquad c_{\ell}^{(r)} = 0 \quad \text{for all other indices.}$$

For the finite computation, ϕ_R is replaced by the polynomial ϕ_A in (4.15). The coefficients $c_{\ell}^{(r,A)}$ and matrices $R_{r;j,k}^{(A)}$ are recomputed from $(a_1,\ldots,a_A)$ through (4.8)–(4.10). We then form $D_{n,J}^{(A)}(\lambda)$ by (4.17) and $W_{N,M,J}^{(A)}(\lambda)$ by (4.18). The same recursions apply to any coefficient vector whose polynomial defines a conformal map on a neighborhood of $\overline{\mathbb{D}}$.

5.2 A Forward Numerical Experiment

We take $R = 1.8$. The computation uses the finite matrix $W_{N,M,J}^{(A)}(\lambda)$ defined in (4.18). Here A truncates the Riemann-map series, N is the retained Fourier half-width, and M, J truncate the outer and inner Taylor sums. The geometry-dependent coefficients $\widehat{q}_{\ell}^{(A)}$, $\mu_s^{(n,A)}$, and $R_{r;j,k}^{(A)}$ are assembled once and cached. Only the scalar Taylor weights and the final matrix assembly are repeated as λ varies. The calculations were implemented in R. Matrix singular values were computed by numerical singular-value decomposition.

To enlarge the four truncations simultaneously, we set

$$N_\ell = 2\ell+2, \qquad A_\ell = 2N_\ell+1, \qquad M_\ell = 2N_\ell+4, \qquad J_\ell = N_\ell+M_\ell, \tag{5.6}$$

for $\ell = 4,5,6,7,8$. The resulting quadruples $(A_\ell, N_\ell, M_\ell, J_\ell)$ are

$$(21,10,24,34), \quad (25,12,28,40), \quad (29,14,32,46), \quad (33,16,36,52), \quad (37,18,40,58).$$

For the map in (5.1), the explicit tail bound for the omitted Riemann-map coefficients decreases from 3.89×10^{-6} at $\ell = 4$ to 3.20×10^{-10} at $\ell = 8$.

For every finite square matrix,

$$\sigma_{\min}\big(W_{N,M,J}^{(A)}(\lambda)\big) = 0 \quad \Longleftrightarrow \quad \det W_{N,M,J}^{(A)}(\lambda) = 0. \tag{5.7}$$

The finite-dimensional regularized determinant in (4.21) differs from the ordinary determinant by a nonzero exponential factor. Hence it has the same zeros. For the numerical search we use the scale-free diagnostic

$$\rho_N(\lambda) \stackrel{\text{def}}{=} \frac{\sigma_{\min}(W^{(A)}_{N,M,J}(\lambda))}{\sigma_{\max}(W^{(A)}_{N,M,J}(\lambda))}. \tag{5.8}$$

Here A, M, J are the cutoffs assigned to the same truncation level as N in (5.6).

We first sample $0.2 \le \lambda \le 25$ with spacing 0.02. Every local minimum of the sampled function

$$\lambda \longmapsto \log_{10} \sigma_{\min}\big(W^{(A)}_{N,M,J}(\lambda)\big)$$

is then refined by one-dimensional minimization with interval tolerance 10^{-10}. A refined minimum is retained as a spectral candidate when

$$\rho_N(\lambda) < 10^{-7}. \tag{5.9}$$

At the finest truncation, all seven candidates satisfy the stronger bound $\rho_N(\lambda) < 3.1 \times 10^{-9}$. Their values are reported in Table 1.

Table 1: The seven retained Wronskian candidates for $R = 1.8$, computed with $(A, N, M, J) = (37, 18, 40, 58)$. The last column contains the coarse finite-element comparison described below.

k	λ_k^W	$\sigma_{\min}$	ρ_N	λ_k^{FEM}
1	4.159287	8.30×10^{-10}	8.56×10^{-10}	4.1667
2	7.328271	8.84×10^{-11}	9.33×10^{-11}	7.3489
3	12.828313	2.71×10^{-9}	2.98×10^{-9}	12.8902
4	13.488701	2.76×10^{-9}	3.05×10^{-9}	13.5564
5	16.819090	3.19×10^{-10}	3.61×10^{-10}	16.9211
6	19.916785	1.11×10^{-9}	1.28×10^{-9}	20.0607
7	23.732970	1.14×10^{-9}	1.36×10^{-9}	23.9361

The displayed digits record the locations returned by the finite minimization and are retained for comparison across truncation levels. They are not claimed as certified digits of the exact Dirichlet eigenvalues.

Let $\lambda_k^{(i)}$ denote the k-th spectral candidate computed at truncation level $\ell = i$. We compare successive spectral locations by

$$\Delta_i^{(k)} \stackrel{\text{def}}{=} \left|\lambda_k^{(i+1)} - \lambda_k^{(i)}\right|, \qquad i = 4, 5, 6, 7. \tag{5.10}$$

This quantity compares the locations of two minima of $\sigma_{\min}$, rather than their attained singular values. For each candidate index, $\Delta_i^{(k)}$ decreases through the levels in (5.6). For the first candidate, $\Delta_4^{(1)}, \dots, \Delta_7^{(1)}$ are

$$9.26 \times 10^{-6}, \quad 9.92 \times 10^{-7}, \quad 1.38 \times 10^{-7}, \quad 6.90 \times 10^{-9},$$

while $\Delta_4^{(7)}, \dots, \Delta_7^{(7)}$ are

$$4.21 \times 10^{-1}, \quad 1.15 \times 10^{-1}, \quad 2.77 \times 10^{-2}, \quad 5.92 \times 10^{-3}.$$

The changes are larger for the higher candidates, while all seven sequences contract through the computed levels, as displayed in Figure 2.

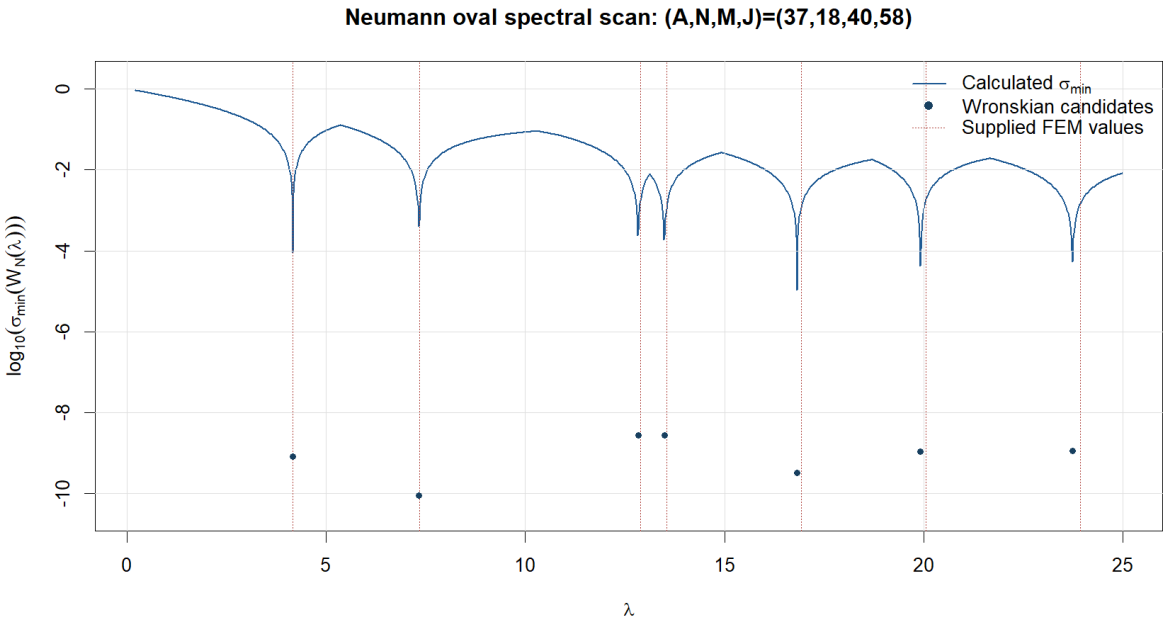


Figure 1: The smallest-singular-value scan at the finest truncation. The solid curve shows $\log_{10}\sigma_{\min}$ on the uniform λ-grid. The dots mark the seven retained Wronskian candidates obtained by refining local minima, and the vertical dashed lines mark the supplied finite-element values.

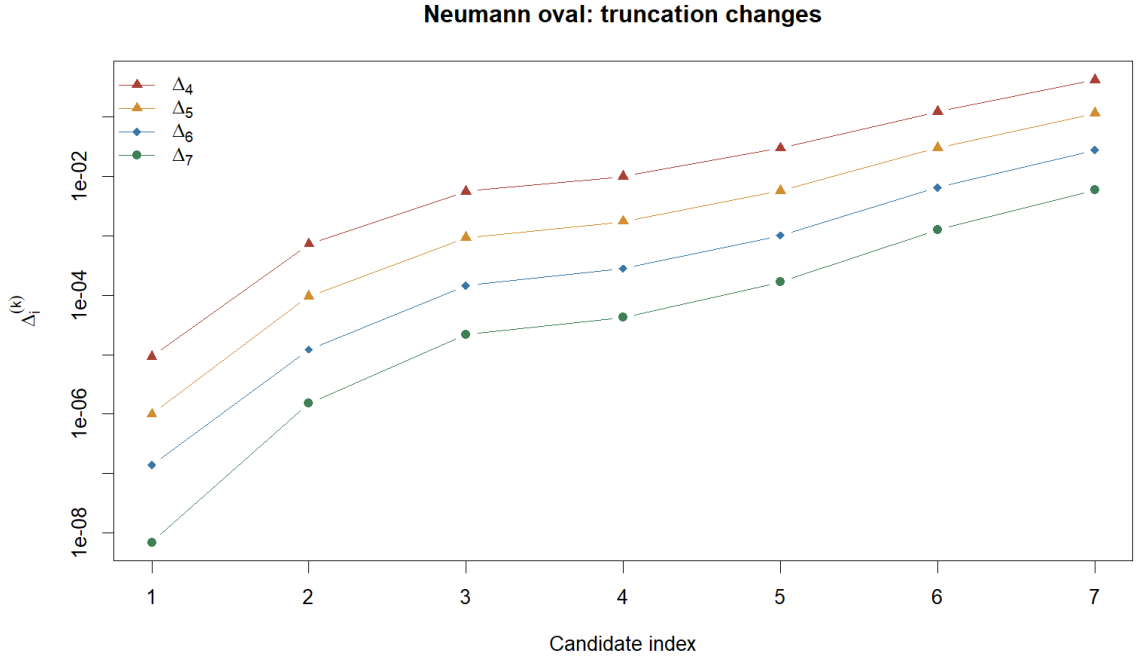


Figure 2: The successive spectral-location changes $\Delta_i^{(k)}$ for the seven retained candidates. Each curve fixes i and varies the candidate index k. The vertical axis is logarithmic.

The comparison values in Table 1 were obtained independently with the MATLAB PDE Toolbox. The Neumann-oval boundary was triangulated by `initmesh`; two uniform refinements were then applied, and piecewise-linear elements were used for the Dirichlet problem. The first 50 values were extracted from the resulting generalized matrix eigenproblem with shift 0.1. Since this calculation was not tested for mesh convergence, the FEM values are used only as a coarse comparison.

We also performed a conformal Galerkin calculation after pulling the eigenvalue problem back to the unit disk. Its trial space consists of Fourier–Bessel functions with angular indices $|m| \le 17$ and 12 radial modes for each index. The integrals containing the weight $|\phi_R'|^2$ were evaluated with 112-point Gauss–Legendre quadrature in the radial variable and a 256-point trapezoidal rule in the angular variable. Comparison with this calculation gives maximum discrepancies over the seven retained values of

$$5.71 \times 10^{-1}, \quad 1.50 \times 10^{-1}, \quad 3.49 \times 10^{-2}, \quad 7.29 \times 10^{-3}, \quad 1.37 \times 10^{-3}$$

for $\ell = 4, 5, 6, 7, 8$, respectively. The discrepancies decrease over the reported truncation range. The finite-element and direct Galerkin calculations therefore provide independent checks on the locations of the low-frequency candidates. Neither is used as a certified high-precision spectrum: the FEM computation lacks a mesh-convergence study, while the Galerkin computation uses a fixed finite trial space and quadrature rule.

5.3 A Restricted Inverse Coefficient-Fitting Experiment

We examine a three-parameter fitting problem for the Neumann oval with $R = 1.8$. At the Riemann-map cutoff $A = 21$, only a_1, a_3, a_5 vary, while $a_7, a_9, \dots, a_{21}$ retain their target values. The trial analytic maps are

$$\phi_{\boldsymbol{\alpha}}(w) = \alpha_1 w + \alpha_3 w^3 + \alpha_5 w^5 + \sum_{m=3}^{10} 2(-1)^m R^{-(2m+1)} w^{2m+1}, \qquad \boldsymbol{\alpha} = (\alpha_1, \alpha_3, \alpha_5) \in \mathbb{R}^3. \tag{5.11}$$

The target values are

$$(a_1, a_3, a_5) = (1.11111111, -0.34293553, 0.10584430).$$

Denote this vector by $\boldsymbol{\alpha}_*$.

Seven synthetic spectral candidates are computed from the target coefficients using $(A, N, M, J) = (21, 10, 10, 14)$. They are

$$\begin{aligned}(\widetilde{\lambda}_1, \dots, \widetilde{\lambda}_7) = (&4.159297, 7.329110, 12.835009, 13.500743,\\ &16.855896, 20.076490, 24.302976).\end{aligned}$$

These values are generated separately from the finer forward candidates in Table 1. For a trial vector $\boldsymbol{\alpha}$, set

$$\rho^{\mathrm{inv}}(\lambda; \boldsymbol{\alpha}) \stackrel{\mathrm{def}}{=} \frac{\sigma_{\min}\big(W^{(21)}_{10,10,14}(\lambda; \boldsymbol{\alpha})\big)}{\sigma_{\max}\big(W^{(21)}_{10,10,14}(\lambda; \boldsymbol{\alpha})\big)}.$$

We minimize

$$\mathcal{F}(\boldsymbol{\alpha}) \stackrel{\mathrm{def}}{=} \sum_{k=1}^{7} \rho^{\mathrm{inv}}(\widetilde{\lambda}_k; \boldsymbol{\alpha})^2. \tag{5.12}$$

The target data and the objective use the same truncated forward model. At the target coefficients, $\mathcal{F}(\boldsymbol{\alpha}_*) = 3.20 \times 10^{-17}$.

The minimization uses the derivative-free Nelder–Mead method [25]. Five runs are initialized at

$$\boldsymbol{\alpha}^{(0)}(\varepsilon) = \big((1+\varepsilon)a_1, (1-\varepsilon)a_3, (1+\varepsilon)a_5\big),$$

where $\varepsilon \in \{0.05, 0.075, 0.10, 0.125, 0.15\}$. We declare a run numerically converged when $\mathcal{F}(\boldsymbol{\alpha}) < 10^{-10}$. In each run, let $s_j = |\alpha_j^{(0)}|$, $z_j = \alpha_j / s_j$, $z_j^{(0)} = \alpha_j^{(0)} / s_j$, and

$$\mathcal{P}_{\boldsymbol{\alpha}^{(0)}} \stackrel{\text{def}}{=} \left\{ \boldsymbol{z} \in \mathbb{R}^3 : |z_j - z_j^{(0)}| \le 0.4,\ j = 1, 3, 5 \right\}.$$

The optimizer is evaluated on the normalized variables $\boldsymbol{z}$. For an inadmissible vector we replace the objective by

$$10^3 \left(1 + \operatorname{dist}(\boldsymbol{z}, \mathcal{P}_{\boldsymbol{\alpha}^{(0)}})^2 + \max\{0, 1 + 10^{-4} - r_{\boldsymbol{\alpha}}\}^2\right),$$

where $r_{\boldsymbol{\alpha}}$ is the smallest modulus of a zero of $\phi'_{\boldsymbol{\alpha}}$. We also require $\alpha_1 > 0$. This condition enforces local conformality near the closed disk, but it does not establish global univalence. The trial vectors are therefore treated as coefficients of analytic maps, and the experiment is a restricted coefficient-fitting test. A geometric domain interpretation requires a separate verification of global injectivity. The R implementation uses `optim` with the derivative-free Nelder–Mead method, relative tolerance 10^{-12}, and a maximum of 300 iterations.

Table 2: Fitted leading coefficients for five initial perturbations. The last two rows report the final objective and the convergence decision.

	Target	5%	7.5%	10%	12.5%	15%
a_1	1.11111111	1.11111116	1.11111126	1.11111121	1.11111109	1.52094929
a_3	−0.34293553	−0.34293550	−0.34293521	−0.34293544	−0.34293555	−0.39827766
a_5	0.10584430	0.10584422	0.10584462	0.10584423	0.10584431	0.17040903
$\mathcal{F}$	—	2.09×10^{-15}	1.63×10^{-14}	5.38×10^{-15}	5.10×10^{-16}	2.37×10^{-4}
Result	—	converged	converged	converged	converged	not converged

The runs initialized between 5% and 12.5% satisfy the residual criterion. Across these four runs, the largest relative error in a recovered coefficient is 3.1×10^{-6}. The 15% run terminates after 301 objective evaluations with $\mathcal{F} = 2.37 \times 10^{-4}$. Thus four of the five tested initial points satisfy the stated criterion in this finite, three-parameter model. Because the targets and the objective share the same forward discretization, the experiment tests local consistency of the coefficient fit. It does not measure sensitivity to independent spectral or truncation errors.

All R calculations used R version 4.6.0 and only base or recommended packages. The source files, parameter configurations, validation scripts, numerical tables, figures, and session information are supplied with the supplementary numerical materials.

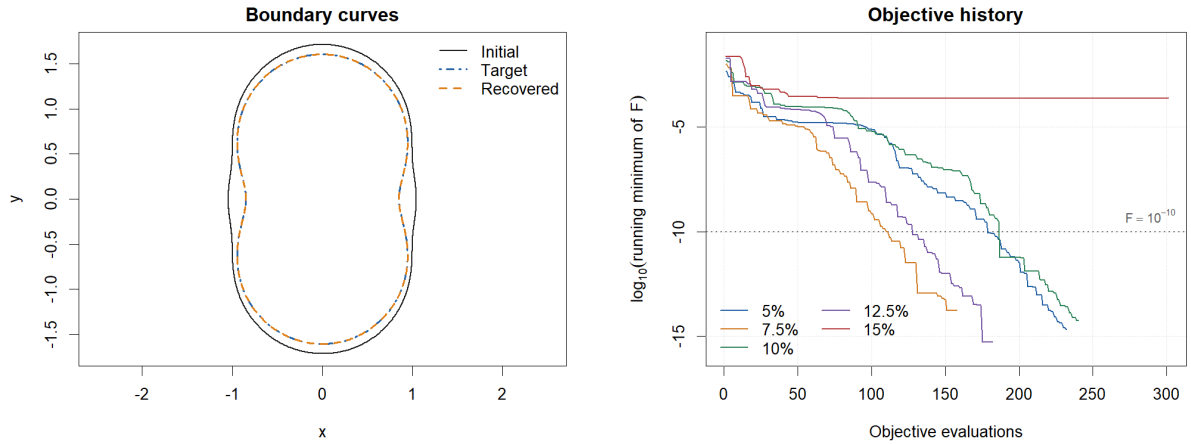


Figure 3: The restricted fitting experiment for a_1, a_3, a_5, with the remaining Neumann coefficients fixed. Left: the target, the 12.5% initial boundary, and the recovered boundary. Right: the running minimum of $\mathcal{F}$ for the five initial perturbations; the horizontal line marks the convergence threshold 10^{-10}.

6 Discussion

In this article, we derive an entire boundary-operator family $W(\lambda) = I + K(\lambda)$ on $\mathcal{H}_\Omega$, where $K(\lambda)$ is Hilbert–Schmidt. For $\lambda > 0$, its kernel is linearly isomorphic to the Dirichlet eigenspace, and $\det{}_2 W(\lambda)$ vanishes exactly at the Dirichlet eigenvalues. Thus $\dim\ker W(\lambda)$ gives the spectral multiplicity, and the regularized determinant gives a scalar spectral criterion. In conformal Fourier coordinates, every matrix entry is determined by the coefficients of the normalized Riemann map.

These formulas yield finite matrices determined by finitely many mapping coefficients. Proposition 4.5 gives convergence of the finite determinants at each fixed λ along coordinated truncations. In the Neumann-oval example, the seven forward candidates stabilize across the reported levels and approach the conformal Galerkin values. The restricted coefficient-fitting experiment varies a_1, a_3, a_5 and fixes the remaining coefficients. Its spectral data are generated by the same finite model, and four of the five tested initializations meet the stated residual criterion. This supports the local computability of the finite model.

Limitations. The present theory does not establish injectivity or stability of the map from Riemann mapping coefficients to spectral data. It identifies spectral multiplicity with $\dim\ker W(\lambda)$. The relation between this multiplicity and the vanishing order of $\det{}_2 W$ remains open. The numerical experiments use a finite set of modes, and the inverse data are generated by the same finite model; they therefore do not provide an independent recovery or stability result.

Further work. Locally uniform truncation estimates on complex λ-neighborhoods, together with contour arguments, may establish convergence of computed zeros. A local expansion of $W(\lambda)$ near an eigenvalue may clarify the corresponding zero order. After fixing the conformal normalization and the interior base point, the coefficient-to-spectrum map may be studied for local injectivity, stability, and finite-data reconstruction. The same harmonic-extension framework may also be adapted to Neumann boundary conditions.

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A Initial Motivation

This appendix records the calculation that first suggested the modal Wronskian. It is included as motivation and is not used in the proof of the main results. Unless a convergence range is stated explicitly, its generating functions, changes of variables, and separated expansions are understood locally or formally; no global well-posedness assertion is attached to these steps. The rigorous construction and spectral equivalence are those of Sections 2 and 3.

Write

$$D_+ = \partial_x + i\partial_y, \qquad D_- = \partial_x - i\partial_y, \qquad D_+D_- = \Delta.$$

Let $z = z(s)$ be a counterclockwise arclength parametrization of $\partial\Omega$, with $z'(s) = e^{i\theta(s)}$, and let ∂_ν denote the outward normal derivative. On the boundary,

$$D_- = e^{-i\theta}(\partial_s + i\partial_\nu), \qquad D_+ = e^{i\theta}(\partial_s - i\partial_\nu).$$

For a solution of (1.1), put

$$p_k = D_-^k u\big|_{\partial\Omega}, \qquad q_k = \partial_\nu D_-^k u\big|_{\partial\Omega}.$$

Then $p_0 = 0$. Applying the two boundary identities to $D_-^k u$ gives

$$p_{k+1} = e^{-i\theta}(p_k' + iq_k), \qquad -\lambda p_{k-1} = e^{i\theta}(p_k' - iq_k).$$

Eliminating q_k, we obtain

$$p_{k+1} = 2e^{-i\theta}p_k' + \lambda e^{-2i\theta}p_{k-1}, \qquad k \geq 1. \tag{A.1}$$

Let S be the Schwarz function of the analytic boundary. Thus $S(z) = \overline{z}$ on $\partial\Omega$, and differentiation along the boundary yields

$$S'(z(s)) = e^{-2i\theta(s)}.$$

The analytic traces p_k extend holomorphically to a boundary collar. Denote these extensions by $\widetilde{W}_k$. Since $\widetilde{W}_k'(z(s)) = e^{-i\theta(s)}p_k'(s)$, the identity theorem extends (A.1) to

$$\widetilde{W}_{k+1} = 2\widetilde{W}_k' + \lambda S'\widetilde{W}_{k-1}, \qquad \widetilde{W}_0 = 0. \tag{A.2}$$

After restricting to a smaller collar, Cauchy's estimates give a number $\rho_0 > 0$ for which the generating function

$$G(\rho, z) = \sum_{k=0}^{\infty} \widetilde{W}_k(z)\frac{(\rho/2)^k}{k!}.$$

converges normally for $|\rho| < \rho_0$. Termwise differentiation there and (A.2) give

$$G_{\rho\rho} = G_{\rho z} + \frac{\lambda}{4}S'(z)G, \qquad G(0, z) = 0. \tag{A.3}$$

Formally set $\rho = (t-1)\eta$, $z = \eta$, and $F(t, \eta) = G((t-1)\eta, \eta)$. Since $z + \rho = t\eta$, $t = 1$ corresponds to the boundary point $z = \eta$ with $\rho = 0$, while $t = 0$ corresponds formally to $z + \rho = 0$. Whenever $|(t-1)\eta| < \rho_0$, direct differentiation gives

$$G_{\rho\rho} = \eta^{-2}F_{tt}, \qquad G_{\rho z} = \eta^{-1}F_{t\eta} - \eta^{-2}F_t - (t-1)\eta^{-2}F_{tt}.$$

Substitution into (A.3) yields the two-point equation

$$tF_{tt} + F_t - \eta F_{t\eta} = \frac{\lambda\eta^2 S'(\eta)}{4} F, \qquad F(1,\eta) = 0. \tag{A.4}$$

To display its separated modes, define

$$\mathfrak{B}_k(x) = \sum_{m=0}^{\infty} \frac{x^m}{m!(m+k)!}, \qquad J_k(r) = \left(\frac{r}{2}\right)^k \mathfrak{B}_k\left(-\frac{r^2}{4}\right).$$

A formal Laurent ansatz in (A.4), followed by comparison of like powers, gives the boundary functions

$$\Phi_p(\eta;\lambda) = \begin{cases} \eta^p \mathfrak{B}_p(-\lambda\eta S(\eta)/4), & p \geq 0, \\ S(\eta)^{-p}\mathfrak{B}_{-p}(-\lambda\eta S(\eta)/4), & p < 0. \end{cases} \tag{A.5}$$

They are the boundary traces of the entire Helmholtz modes

$$\mathcal{U}_p(z,\overline{z};\lambda) = \left(\frac{2}{\sqrt{\lambda}}\right)^{|p|} J_{|p|}(\sqrt{\lambda}|z|) e^{ip\arg z}.$$

For a disk centered at the origin, the standard separation-of-variables calculation gives a local expansion of every analytic Helmholtz solution in these regular Bessel modes. Analyticity excludes the singular radial solutions, while uniqueness of Fourier coefficients fixes the modal coefficients. This observation motivates the corresponding boundary-mode construction; no such local completeness statement is needed in the main proof.

The boundary functions Φ_p are traces of exact Helmholtz solutions. Their coordinate columns in a fixed boundary Hilbert basis form a modal boundary matrix. A nonzero kernel vector records a linear dependence among these traces, which is the sense in which the operator constructed in Section 3 plays the role of a Wronskian. The harmonic-extension construction provides the Hilbert-space realization used in the main argument.

B Supplementary Calculations

B.1 Calculations for Section 3

For Proposition 3.2. Put $F_0 = f \circ \phi$ and $F_r = (T_\Omega^r f) \circ \phi$. Then

$$F_r(w) = \int_0^w \phi'(\zeta) F_{r-1}(\zeta)\, d\zeta, \qquad F_r' = \phi' F_{r-1}.$$

Since the primitive on $H^2(\mathbb{D})$ has norm one,

$$\|F_r\|_{H^2} \leq \|\phi'\|_\infty \|F_{r-1}\|_{H^2} \leq \|\phi'\|_\infty^r \|F_0\|_{H^2},$$

and

$$\|F_r'\|_{H^2} \leq \|\phi'\|_\infty \|F_{r-1}\|_{H^2} \leq \|\phi'\|_\infty^r \|F_0\|_{H^2}.$$

Equation (2.5) therefore gives

$$\|F_r\|_{H^1(\partial\mathbb{D})}^2 = \|F_r\|_{H^2}^2 + \|F_r'\|_{H^2}^2 \leq 2\|\phi'\|_\infty^{2r} \|f\|_{H^2_{\mathrm{hol}}(\Omega)}^2.$$

For $T_{\mathbb{D}} w^k = w^{k+1}/(k+1)$,

$$\|T_{\mathbb{D}}\|_{\mathcal{S}_2}^2 = \sum_{k=0}^{\infty} \frac{1}{(k+1)^2} = \frac{\pi^2}{6}, \qquad \|T_\Omega\|_{\mathcal{S}_2} \leq \frac{\pi}{\sqrt{6}} \|\phi'\|_\infty.$$

Together with $\|T_\Omega\| \le \|\phi'\|_\infty$, the ideal inequality yields

$$\|T_\Omega^r\|_{\mathcal{S}_2} \le \|T_\Omega\|^{r-1}\|T_\Omega\|_{\mathcal{S}_2} \le \frac{\pi}{\sqrt{6}}\|\phi'\|_\infty^r.$$

For Proposition 3.3. Let $K_1 \Subset \Omega$, $K_2 \Subset \Omega^*$, and set

$$\rho_1 = \max_{\eta\in K_1} |\psi(\eta)| < 1, \qquad \rho_2 = \max_{\zeta\in K_2} |\psi^\#(\zeta)| < 1.$$

The Hardy pointwise estimate and the preceding bounds imply

$$|T_\Omega^r f_b(\eta)| \le \frac{\|\phi'\|_\infty^r}{\sqrt{1-\rho_1^2}}\|f_b\|_{H^2_{\mathrm{hol}}},$$

and

$$|(T_\Omega^r g_b)^\#(\zeta)| \le \frac{\|\phi'\|_\infty^r}{\sqrt{1-\rho_2^2}}\|g_b\|_{H^2_{\mathrm{hol}}}.$$

Writing

$$R_1 = \max_{\eta\in K_1} |\eta|, \qquad R_2 = \max_{\zeta\in K_2} |\zeta|,$$

we obtain, for some $C_{K_1,K_2} > 0$,

$$\left|\zeta^r T_\Omega^r f_b(\eta) + \eta^r (T_\Omega^r g_b)^\#(\zeta)\right| \le C_{K_1,K_2}\big(\|\phi'\|_\infty \max\{R_1,R_2\}\big)^r \|b\|_{\mathcal{H}_\Omega}.$$

Consequently, for $|\lambda| \le \Lambda$, the absolute majorant is

$$C_{K_1,K_2}\|b\|_{\mathcal{H}_\Omega} \sum_{r=0}^{\infty} \frac{(\Lambda\|\phi'\|_\infty \max\{R_1,R_2\}/4)^r}{r!},$$

which is finite. Applying the same estimate on slightly larger compact sets and using the Cauchy inequalities proves normal convergence of every derivative series.

For the first half of (3.7), normal convergence permits the calculation

$$\begin{aligned}\partial_\eta\partial_{\overline{\eta}} \sum_{r=0}^{\infty} \frac{(-\lambda/4)^r}{r!}\overline{\eta}^r T_\Omega^r f_b &= \sum_{r=1}^{\infty} \frac{(-\lambda/4)^r}{(r-1)!}\overline{\eta}^{r-1} T_\Omega^{r-1} f_b \\ &= -\frac{\lambda}{4}\sum_{s=0}^{\infty} \frac{(-\lambda/4)^s}{s!}\overline{\eta}^s T_\Omega^s f_b.\end{aligned}$$

The reflected half is identical after interchanging η and $\overline{\eta}$. Thus $4\partial_\eta\partial_{\overline{\eta}} u_b = -\lambda u_b$.

For the radial trace, write $F_r = (T_\Omega^r f_b)\circ\phi$ and $G_r = (T_\Omega^r g_b)\circ\phi$. The maximum principle and the Hardy norm bounds give

$$\sup_{0<\rho<1} \|\overline{\phi(\rho e^{i\theta})}^r F_r(\rho e^{i\theta})\|_{L^2} \le (L\|\phi'\|_\infty)^r \|f_b\|_{H^2_{\mathrm{hol}}},$$

with the same estimate for the reflected term. Hence

$$\sup_{0<\rho<1} \sum_{r>R} \frac{(|\lambda|/4)^r}{r!}\|V_{r,\rho}\|_{L^2} \le \sqrt{2}\|b\|_{\mathcal{H}_\Omega} \sum_{r>R} \frac{(|\lambda|L\|\phi'\|_\infty/4)^r}{r!} \longrightarrow 0.$$

Each of the first R terms has a radial L^2 limit. Taking that finite limit and then letting $R \to \infty$ proves that the limit may be passed through the series.

The complexification satisfies

$$U_b(\eta,0;\lambda) = f_b(\eta), \qquad U_b(0,\zeta;\lambda) = f_b(0) + g_b^\#(\zeta),$$

because $T_\Omega^r f_b$ and $T_\Omega^r g_b$ vanish to order at least r. If $u_b \equiv 0$, all Wirtinger derivatives at the origin vanish, and therefore

$$f_b^{(k)}(0) = 0 \quad (k \ge 0), \qquad (g_b^\#)^{(k)}(0) = 0 \quad (k \ge 1).$$

The identity theorem gives $f_b = g_b = 0$, and hence $b = 0$. If b is replaced by αb, then $f_{\alpha b} = \alpha f_b$ and

$g_{\alpha b} = \overline{\alpha} g_b$; the final conjugation in the second half of (3.7) restores the factor α. This verifies complex linearity.

For Theorem 3.6. For $V = U - U_b$, write

$$V(\eta, \zeta) = \sum_{i,j \geq 0} \delta_{ij} \eta^i \zeta^j.$$

The two axis identities give $\delta_{i0} = \delta_{0j} = 0$. Since $V_{\eta\zeta} = -\lambda V/4$, coefficient comparison gives, for $i, j \geq 1$,

$$ij\delta_{ij} = -\frac{\lambda}{4} \delta_{i-1,j-1}.$$

Let $m = \min\{i, j\}$. Iteration yields

$$\delta_{ij} = \left(-\frac{\lambda}{4}\right)^m \frac{(i-m)!(j-m)!}{i!j!} \delta_{i-m,j-m} = 0,$$

because $i - m = 0$ or $j - m = 0$. Thus $U = U_b$ near the origin, and successive one-variable analytic continuation gives equality on $\Omega \times \Omega^*$.

For Lemma 3.7. Termwise differentiation gives

$$\begin{aligned} \mathfrak{B}_k^{(n)}(x) &= \sum_{m=n}^{\infty} \frac{m!}{(m-n)!} \frac{x^{m-n}}{m!(m+k)!} \\ &= \sum_{\ell=0}^{\infty} \frac{x^\ell}{\ell!(\ell+k+n)!} = \mathfrak{B}_{k+n}(x). \end{aligned}$$

Furthermore,

$$\begin{aligned} \sum_{r=0}^{\infty} \frac{(-\lambda/4)^r}{r!} \overline{\eta}^r T_\Omega^r \left(\frac{\eta^k}{k!}\right) &= \sum_{r=0}^{\infty} \frac{(-\lambda/4)^r}{r!} \frac{\eta^{k+r} \overline{\eta}^r}{(k+r)!} \\ &= \eta^k \mathfrak{B}_k \left(-\frac{\lambda|\eta|^2}{4}\right). \end{aligned}$$

For Proposition 3.8. The unitary pullback of H is multiplication by $q(e^{i\theta}) = |\phi(e^{i\theta})|^2$. Therefore

$$\|Hb\|_{\mathcal{H}_\Omega}^2 = \frac{1}{2\pi} \int_0^{2\pi} q(e^{i\theta})^2 |b(\phi(e^{i\theta}))|^2 \, d\theta,$$

and $d^2 \leq q \leq L^2$ gives $\|H\| = L^2$ and $\|H^{-1}\| = d^{-2}$. For $k, j \in \mathbb{Z}$,

$$\begin{aligned} \langle HE_k, E_j \rangle_{\mathcal{H}_\Omega} &= \frac{1}{2\pi} \int_0^{2\pi} q(e^{i\theta}) e^{ik\theta} e^{-ij\theta} \, d\theta \\ &= \widehat{q}_{j-k}. \end{aligned}$$

If q is holomorphic on $\beta^{-1} < |w| < \beta$, Laurent's formula gives

$$|\widehat{q}_\ell| \leq \max_{|w|=\beta} |q(w)| \, \beta^{-\ell} \quad (\ell \geq 0),$$

and

$$|\widehat{q}_{-\ell}| \leq \max_{|w|=\beta^{-1}} |q(w)| \, \beta^{-\ell} \quad (\ell \geq 0).$$

For Proposition 3.10. Set $\chi(w) = \phi(w)/w$, with $\chi(0) = \phi'(0)$. Division by w^r is unitary from $w^r H^2(\mathbb{D})$ onto $H^2(\mathbb{D})$. Since χ is nonzero on $\overline{\mathbb{D}}$, the minimum-modulus principle gives

$$\min_{|w| \leq 1} |\chi(w)| = \min_{|w|=1} |\chi(w)| = \min_{|w|=1} |\phi(w)| = d.$$

Thus

$$(\mathcal{R}_r^+ f)\circ\phi = \chi^{-r}\frac{(T_\Omega^r f)\circ\phi}{w^r}$$

and

$$\|\mathcal{R}_r^+\|_{\mathcal{S}_2} \le \|\chi^{-r}\|_\infty \|T_\Omega^r\|_{\mathcal{S}_2} \le \frac{\pi}{\sqrt{6}}\left(\frac{\|\phi'\|_\infty}{d}\right)^r.$$

For $k \ge 1$, E_k contributes through $f_{E_k} = \psi^k$, whereas E_{-k} contributes through $g_{E_{-k}} = \psi^k$. The two blocks have equal squared norms, and $\mathcal{R}_r E_0 = E_0/r!$. Hence

$$\begin{aligned}\|\mathcal{R}_r\|_{\mathcal{S}_2}^2 &= \frac{1}{(r!)^2} + \sum_{k=1}^{\infty}\|\mathcal{R}_r^+(\psi^k)\|_{H^2_{\mathrm{hol}}}^2 + \sum_{k=1}^{\infty}\|\mathcal{R}_r^+(\psi^k)\|_{H^2_{\mathrm{hol}}}^2\\ &\le \frac{1}{(r!)^2} + 2\|\mathcal{R}_r^+\|_{\mathcal{S}_2}^2,\end{aligned}$$

which is (3.18). Direct integration also gives

$$\eta^{-r}T_\Omega^r(\eta^k) = \frac{k!}{(k+r)!}\eta^k.$$

On the boundary, reflection gives the same coefficient for S^k.

For Proposition 3.11. For $n \ge 1$, the partial sums satisfy

$$\begin{aligned}\sum_{m=0}^{J}\left\|\frac{(-\lambda\widehat{q}_0/4)^m}{m!}\mathcal{R}_{n+m}\right\|_{\mathcal{S}_2} &\le C_0 C_\Omega^n \sum_{m=0}^{J}\frac{(|\lambda|\widehat{q}_0 C_\Omega/4)^m}{m!}\\ &\le C_0 C_\Omega^n \exp\left(\frac{|\lambda|\widehat{q}_0 C_\Omega}{4}\right).\end{aligned}$$

For $n = 0$, isolate the term $m = 0$, which equals I, and obtain

$$\sum_{m=1}^{\infty}\left\|\frac{(-\lambda\widehat{q}_0/4)^m}{m!}\mathcal{R}_m\right\|_{\mathcal{S}_2} \le C_0\left[\exp\left(\frac{|\lambda|\widehat{q}_0 C_\Omega}{4}\right) - 1\right].$$

Both majorants are uniform on compact subsets of the λ-plane. Completeness of $\mathcal{S}_2$ and the Weierstrass theorem for Banach-valued power series give the claimed entire dependence. Finally, with $k = |p|$,

$$\begin{aligned}D_n(\lambda)\beta_p &= \sum_{m=0}^{\infty}\frac{(-\lambda\widehat{q}_0/4)^m}{m!}\frac{k!}{(k+n+m)!}\beta_p\\ &= k!\,\mathfrak{B}_{k+n}\left(-\frac{\lambda\widehat{q}_0}{4}\right)\beta_p.\end{aligned}$$

For Proposition 3.12. After substituting (3.22), the sum of the Hilbert–Schmidt norms of all terms except the identity is bounded by

$$\begin{aligned}&C_0\sum_{\substack{n,m\ge 0\\ n+m\ge 1}}\frac{(|\lambda|C_\Omega\|H-\widehat{q}_0 I\|/4)^n}{n!}\frac{(|\lambda|C_\Omega\widehat{q}_0/4)^m}{m!}\\ &\quad= C_0\left[\exp\left(\frac{|\lambda|C_\Omega}{4}(\|H-\widehat{q}_0 I\|+\widehat{q}_0)\right)-1\right].\end{aligned}$$

The double series may therefore be grouped by $r = n + m$. For fixed r,

$$\begin{aligned}
&\sum_{n=0}^{r} \frac{(-\lambda/4)^n}{n!}(H - \widehat{q}_0 I)^n \frac{(-\lambda\widehat{q}_0/4)^{r-n}}{(r-n)!}\mathcal{R}_r \\
&\quad = \frac{(-\lambda/4)^r}{r!}\sum_{n=0}^{r}\binom{r}{n}(H - \widehat{q}_0 I)^n(\widehat{q}_0 I)^{r-n}\mathcal{R}_r \\
&\quad = \frac{(-\lambda/4)^r}{r!}H^r\mathcal{R}_r.
\end{aligned}$$

Equation (3.20) identifies the last series with the radial trace defining $W(\lambda)$. Moreover,

$$\begin{aligned}
\|W(\lambda) - I\|_{\mathcal{S}_2} &\le \sum_{r=1}^{\infty} \frac{(|\lambda|/4)^r}{r!}\|H\|^r\|\mathcal{R}_r\|_{\mathcal{S}_2} \\
&\le C_0 \sum_{r=1}^{\infty} \frac{(|\lambda|L^2 C_\Omega/4)^r}{r!} \\
&= C_0\left[e^{|\lambda|L^2C_\Omega/4} - 1\right].
\end{aligned}$$

For Proposition 3.14. Let $G_r = T_\Omega^r f$ and $F_r = G_r \circ \phi$. Change of variables and conformal invariance of the Dirichlet integral give

$$\|G_r\|_{L^2(\Omega)}^2 \le \pi\|\phi'\|_\infty^2\|F_r\|_{H^2}^2 \le \pi\|\phi'\|_\infty^{2r+2}\|f\|_{H^2_{\mathrm{hol}}}^2,$$

and

$$\begin{aligned}
\|\nabla G_r\|_{L^2(\Omega)}^2 &= \|\nabla F_r\|_{L^2(\mathbb{D})}^2 \\
&= 2\pi\sum_{k\ge 1} k|\widehat{F}_r(k)|^2 \le 2\pi\|F_r'\|_{H^2}^2 \\
&\le 2\pi\|\phi'\|_\infty^{2r}\|f\|_{H^2_{\mathrm{hol}}}^2.
\end{aligned}$$

Thus, with $C_1 = \sqrt{\pi}(\|\phi'\|_\infty + \sqrt{2})$,

$$\|G_r\|_{H^1(\Omega)} \le C_1\|\phi'\|_\infty^r\|f\|_{H^2_{\mathrm{hol}}}.$$

The product rule yields

$$\begin{aligned}
\|\overline{\eta}^r G_r\|_{L^2(\Omega)} &\le L^r\|G_r\|_{L^2(\Omega)}, \\
\|\nabla(\overline{\eta}^r G_r)\|_{L^2(\Omega)} &\le L^r\|\nabla G_r\|_{L^2(\Omega)} + \sqrt{2}rL^{r-1}\|G_r\|_{L^2(\Omega)}.
\end{aligned}$$

On $\partial\mathbb{D}$,

$$\frac{d}{d\theta}\left(\overline{\phi(e^{i\theta})}^r F_r(e^{i\theta})\right) = r\overline{\phi}^{r-1}\frac{d}{d\theta}\overline{\phi}\,F_r + \overline{\phi}^r\frac{d}{d\theta}F_r,$$

and consequently

$$\begin{aligned}
\|\overline{\phi}^r F_r\|_{L^2} &\le L^r\|F_r\|_{H^2}, \\
\left\|\frac{d}{d\theta}(\overline{\phi}^r F_r)\right\|_{L^2} &\le rL^{r-1}\|\phi'\|_\infty\|F_r\|_{H^2} + L^r\|F_r'\|_{H^2}.
\end{aligned}$$

The bounds for F_r and F_r', the equivalent boundary norms under ϕ, and the identical reflected estimates produce constants $C_2 > 0$, $A_\Omega \ge 1$ such that

$$\|v_{r,b}\|_{H^1(\Omega)} + \|v_{r,b}|_{\partial\Omega}\|_{H^1(\partial\Omega)} \le C_2(1+r)A_\Omega^r\|b\|_{\mathcal{H}_\Omega}.$$

The factorial sum used for the correction is

$$\sum_{r=1}^{\infty}\frac{x^r(1+r)}{r!} = (e^x - 1) + xe^x.$$

For Theorem 3.15. If $b \in \ker W(\lambda)$, then $b = -K(\lambda)b \in H^1(\partial\Omega) \subset H^{1/2}(\partial\Omega)$. Proposition 3.14 therefore gives

$$u_b \in H^1(\Omega), \qquad \gamma u_b = W(\lambda)b = 0, \qquad u_b \in H_0^1(\Omega).$$

The pointwise equation also holds distributionally. Hence, for $v \in C_c^\infty(\Omega)$,

$$\int_\Omega \nabla u_b \cdot \nabla \overline{v}\, dA = \lambda \int_\Omega u_b \overline{v}\, dA.$$

Density of $C_c^\infty(\Omega)$ in $H_0^1(\Omega)$ extends this identity to every $v \in H_0^1(\Omega)$. Conversely, if $u \in E_\lambda^D$, Theorem 3.6 gives a unique $b \in H^{1/2}(\partial\Omega)$ with $u = u_b$, and

$$W(\lambda)b = \gamma u = 0.$$

The two assignments are inverse because Proposition 3.3 makes $b \mapsto u_b$ injective.

For Corollary 3.17. Take $v = u_b$ in the weak equation. Then

$$\lambda = \frac{\displaystyle\int_\Omega |\nabla u_b|^2\, dA}{\displaystyle\int_\Omega |u_b|^2\, dA} > 0.$$

Both integrals are real, and the numerator can vanish only when $u_b = 0$, which has already been excluded.

B.2 Calculations for Section 4

For Proposition 4.1. Fix $1 < \tau < R_\phi$. Cauchy's estimate gives $|a_j| \le C_\tau \tau^{-j}$, and hence

$$\sum_{j,m\ge 1} |a_j \overline{a_m}| \le C_\tau^2 \left(\sum_{j\ge 1} \tau^{-j} \right)^2 < \infty.$$

Thus the product may be rearranged:

$$\begin{aligned} q(e^{i\theta}) &= \left(\sum_{j\ge 1} a_j e^{ij\theta} \right) \left(\sum_{m\ge 1} \overline{a_m} e^{-im\theta} \right) \\ &= \sum_{\ell\in\mathbb{Z}} \left(\sum_{m\ge \max\{1,1-\ell\}} a_{m+\ell}\overline{a_m} \right) e^{i\ell\theta}. \end{aligned}$$

For $\ell \ge 0$, this gives

$$\widehat{q}_\ell = \sum_{m\ge 1} a_{m+\ell}\overline{a_m}, \qquad \sum_{m\ge 1} |a_{m+\ell} a_m| \le \frac{C_\tau^2}{\tau^2 - 1} \tau^{-\ell}.$$

For Proposition 4.2. The coefficient of w^m in $\mathcal{T}_\phi F$ is

$$\begin{aligned} [w^m]\mathcal{T}_\phi F &= \sum_{\substack{n\ge 0,\ j\ge 1 \\ n+j=m}} \frac{j a_j}{n+j} x_n \\ &= \sum_{n=0}^{m-1} \frac{m-n}{m} a_{m-n} x_n. \end{aligned}$$

The first rows of the matrix are

$$[\mathcal{T}_\phi] = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots \\ a_1 & 0 & 0 & 0 & \cdots \\ a_2 & \frac{a_1}{2} & 0 & 0 & \cdots \\ a_3 & \frac{2a_2}{3} & \frac{a_1}{3} & 0 & \cdots \\ a_4 & \frac{3a_3}{4} & \frac{a_2}{2} & \frac{a_1}{4} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

If $F\phi' = \sum_{m\ge 0} g_m w^m$, then

$$\|\mathcal{T}_\phi F\|_{H^2}^2 = \sum_{m\ge 0} \frac{|g_m|^2}{(m+1)^2} \le \sum_{m\ge 0} |g_m|^2 \le \|\phi'\|_\infty^2 \|F\|_{H^2}^2.$$

For equations (4.8)–(4.11)**.** From

$$\phi'(w) F_{r,k}(w) = \sum_{j\ge 1}\sum_{h\ge 0} j a_j t_{h,k}^{(r)} w^{j+h-1},$$

the coefficient of $w^{\ell-1}$ is $\sum_{j=1}^{\ell} j a_j t_{\ell-j,k}^{(r)}$. Integration divides this coefficient by ℓ, which gives

$$t_{\ell,k}^{(r+1)} = \frac{1}{\ell} \sum_{j=1}^{\ell} j a_j t_{\ell-j,k}^{(r)}.$$

Since $F_{r,k}$ vanishes to order at least $k+r$,

$$\begin{aligned} \frac{F_{r,k}(w)}{\phi(w)^r} &= \left(\frac{w}{\phi(w)}\right)^r \frac{F_{r,k}(w)}{w^r} \\ &= \left(\sum_{j\ge 0} c_j^{(r)} w^j\right)\left(\sum_{h\ge 0} t_{h+r,k}^{(r)} w^h\right). \end{aligned}$$

Therefore

$$s_{\ell,k}^{(r)} = \sum_{j=0}^{\ell} c_j^{(r)} t_{\ell-j+r,k}^{(r)}.$$

Writing $\phi(w)/w = \sum_{j\ge 0} a_{j+1} w^j$, the identity

$$\left(\sum_{j\ge 0} a_{j+1} w^j\right)\left(\sum_{h\ge 0} c_h^{(1)} w^h\right) = 1$$

gives

$$a_1 c_0^{(1)} = 1, \qquad a_1 c_\ell^{(1)} + \sum_{j=1}^{\ell} a_{j+1} c_{\ell-j}^{(1)} = 0 \quad (\ell \ge 1).$$

The coefficients of $\chi^{-r} = (\chi^{-1})^r$ are obtained by convolution. For the leading term, use

$$\psi(\eta)^k = a_1^{-k}\eta^k + O(\eta^{k+1}),$$

which gives

$$T_\Omega^r(\psi^k)(\eta) = a_1^{-k} \frac{k!}{(k+r)!} \eta^{k+r} + O(\eta^{k+r+1}).$$

After division by η^r and composition with $\phi(w) = a_1 w + O(w^2)$,

$$\frac{T_\Omega^r(\psi^k)(\phi(w))}{\phi(w)^r} = \frac{k!}{(k+r)!} w^k + O(w^{k+1}).$$

Thus $s^{(r)}_{\ell,k} = 0$ for $\ell < k$ and $s^{(r)}_{k,k} = k!/(k+r)!$. The negative-frequency block is its coefficientwise conjugate, while all mixed blocks vanish.

For Proposition 4.4. Taking matrix entries in (4.12) gives

$$[D_n(\lambda)]_{\widetilde{B};j,k} = \sum_{m=0}^{\infty} \frac{(-\lambda \widehat{q}_0/4)^m}{m!} [\mathcal{R}_{n+m}]_{\widetilde{B};j,k}.$$

Since multiplication of operator matrices corresponds to summation over the intermediate Fourier index,

$$\begin{aligned}[W(\lambda)]_{\widetilde{B};j,k} &= \sum_{n=0}^{\infty} \frac{(-\lambda/4)^n}{n!} \sum_{\ell\in\mathbb{Z}} \widehat{(q-\widehat{q}_0)^n}_{\,j-\ell} [D_n(\lambda)]_{\widetilde{B};\ell,k} \\ &= \sum_{r=0}^{\infty} \frac{(-\lambda/4)^r}{r!} \sum_{\ell\in\mathbb{Z}} \widehat{q^r}_{\,j-\ell} [\mathcal{R}_r]_{\widetilde{B};\ell,k}.\end{aligned}$$

For each fixed j, k, the inner sums are absolutely convergent by Cauchy–Schwarz, since a row of a bounded Laurent multiplier lies in ℓ^2 and a column of a Hilbert–Schmidt operator lies in ℓ^2.

For the coefficient cutoff (4.15)**.** For $1 < \tau < R_\phi$,

$$\|\phi - \phi_A\|_{L^\infty(\overline{\mathbb{D}})} \le C_\tau \sum_{j>A} \tau^{-j} = \frac{C_\tau \tau^{-A}}{\tau - 1},$$

and, with $x = \tau^{-1}$,

$$\begin{aligned}\|\phi' - \phi'_A\|_{L^\infty(\overline{\mathbb{D}})} &\le C_\tau \sum_{j=A+1}^{\infty} j\tau^{-j} \\ &= C_\tau \frac{x^{A+1}((A+1) - Ax)}{(1-x)^2} \longrightarrow 0.\end{aligned}$$

It follows that

$$\|q_A - q\|_\infty \le (\|\phi_A\|_\infty + \|\phi\|_\infty)\|\phi_A - \phi\|_\infty \longrightarrow 0.$$

For $\chi_A = \phi_A / w$, once $\|\chi_A - \chi\|_\infty \le \delta_\phi/2$,

$$\inf_{\overline{\mathbb{D}}} |\chi_A| \ge \frac{\delta_\phi}{2}, \qquad \|\chi_A^{-1} - \chi^{-1}\|_\infty \le \frac{2}{\delta_\phi^2} \|\chi_A - \chi\|_\infty.$$

For fixed r, the telescoping identity

$$\chi_A^{-r} - \chi^{-r} = (\chi_A^{-1} - \chi^{-1}) \sum_{h=0}^{r-1} \chi_A^{-h} \chi^{-(r-1-h)}$$

proves uniform convergence of the reciprocal powers.

For the tail bounds (4.19) **and** (4.20)**.** The first remainder satisfies

$$\begin{aligned}&\left\| D_n(\lambda) - \sum_{m=0}^{J} \frac{(-\lambda \widehat{q}_0/4)^m}{m!} \mathcal{R}_{n+m} \right\|_{\mathcal{S}_2} \\ &\le \sum_{m>J} \frac{(|\lambda| \widehat{q}_0/4)^m}{m!} \|\mathcal{R}_{n+m}\|_{\mathcal{S}_2} \\ &\le C_0 C_\Omega^n \sum_{m>J} \frac{(|\lambda| \widehat{q}_0 C_\Omega/4)^m}{m!}.\end{aligned}$$

Similarly,

$$\begin{aligned}
&\left\| W(\lambda) - \sum_{n=0}^{M} \frac{(-\lambda/4)^n}{n!} (H - \widehat{q}_0 I)^n D_n(\lambda) \right\|_{\mathcal{S}_2} \\
&\quad \le \sum_{n>M} \frac{(|\lambda|/4)^n}{n!} \|H - \widehat{q}_0 I\|^n \|D_n(\lambda)\|_{\mathcal{S}_2} \\
&\quad \le C_0 e^{|\lambda| \widehat{q}_0 C_\Omega/4} \sum_{n>M} \frac{(|\lambda| C_\Omega \|q - \widehat{q}_0\|_\infty/4)^n}{n!}.
\end{aligned}$$

For Proposition 4.5. The Hilbert–Schmidt estimate for the disk primitive and the ideal inequality give

$$\|\mathcal{T}_{\phi_A} - \mathcal{T}_\phi\|_{\mathcal{S}_2} \le \frac{\pi}{\sqrt{6}} \|\phi'_A - \phi'\|_\infty \longrightarrow 0.$$

For fixed r,

$$\mathcal{T}_{\phi_A}^r - \mathcal{T}_\phi^r = \sum_{h=0}^{r-1} \mathcal{T}_{\phi_A}^h (\mathcal{T}_{\phi_A} - \mathcal{T}_\phi) \mathcal{T}_\phi^{r-1-h},$$

so boundedness of the surrounding factors and the ideal inequality give convergence in $\mathcal{S}_2$.

Division by w^r is an isometry on the ranges of these powers. Consequently, the positive-frequency blocks satisfy

$$\begin{aligned}
\|\mathbf{R}_{r,+}^{(A)} - \mathbf{R}_{r,+}\|_{\mathcal{S}_2} \le \|\chi_A^{-r}\|_\infty \|\mathcal{T}_{\phi_A}^r - \mathcal{T}_\phi^r\|_{\mathcal{S}_2} \\
+ \|\chi_A^{-r} - \chi^{-r}\|_\infty \|\mathcal{T}_\phi^r\|_{\mathcal{S}_2} \longrightarrow 0.
\end{aligned}$$

Reflection gives the same conclusion for the negative block, while the constant column is independent of A. Hence

$$\|\mathbf{R}_r^{(A)} - [\mathcal{R}_r]_{\widetilde{B}}\|_{\mathcal{S}_2} \longrightarrow 0.$$

For fixed n, J, it follows term by term that

$$\|D_{n,J}^{(A)} - D_{n,J}\|_{\mathcal{S}_2} \longrightarrow 0.$$

Write $M_{H,A}$ for the Laurent multiplication matrix with symbol q_A. Since its operator norm equals the L^∞-norm of the symbol, uniform convergence of q_A gives, for fixed n,

$$\|(M_{H,A} - \widehat{q}_0^{(A)} I)^n - (M_H - \widehat{q}_0 I)^n\| \longrightarrow 0.$$

The finite sum over $n \le M$, followed by compression to I_N, yields

$$\begin{aligned}
&\|W_{N,M,J}^{(A)} - W_{N,M,J}^{(\infty)}\|_{\mathrm{F}} \\
&\quad \le \left\| \sum_{n=0}^{M} \frac{(-\lambda/4)^n}{n!} \left[(M_{H,A} - \widehat{q}_0^{(A)} I)^n D_{n,J}^{(A)} - (M_H - \widehat{q}_0 I)^n D_{n,J} \right] \right\|_{\mathcal{S}_2} \longrightarrow 0.
\end{aligned}$$

Finite-dimensional continuity of $B \mapsto \det B \exp[-\operatorname{tr}(B-I)]$ gives the associated determinant convergence.

Finally, let $\varepsilon_\nu \downarrow 0$. First choose $N_\nu \ge \nu$ so that

$$\|K - \Pi_{N_\nu} K \Pi_{N_\nu}\|_{\mathcal{S}_2} < \varepsilon_\nu.$$

Next choose $M_\nu, J_\nu \ge \nu$ so that the two factorial tails are below ε_ν, and then choose $A_\nu \ge \max\{A_0, \nu\}$ so that the fixed-block error is below ε_ν. The triangle inequality gives convergence in the finite-dimensional Hilbert–Schmidt norm, and continuity of $\det_2$ proves (4.24).

Acknowledgments

(1) Background, supervision, and assistance from others

This project grew from the author's interest in how sound may distinguish different objects or musical instruments. Through discussions with Jian Deng, this curiosity developed into a question in inverse spectral geometry. After studying the development of the problem, the author focused on planar domains with real-analytic boundaries.

The relationship between the author and Jian Deng is that of a student and an external academic advisor. Jian Deng suggested the research topic, advised the author on its mathematical scope, and provided guidance on the theoretical construction, numerical formulation, and presentation of the paper. He also pointed out gaps in several operator arguments and helped clarify the representation result required by the spectral construction.

The relationship between the author and Lingyun Chang is that of a student and a school teacher. Lingyun Chang advised the author on academic writing, clarity of exposition, organization of the report, presentation of references, and academic conventions.

Both advisors provided their guidance without payment. Except for the assistance described above, no other person or institution carried out the mathematical derivations, numerical experiments, or preparation of this report on behalf of the author.

(2) Research process and contributions

1. **Initial algebraic construction.** The author first studied Lie-group and Lie-algebra actions on solutions of the Helmholtz equation. Certain differential operators led to formulas involving iterated primitive operators and normalized Bessel coefficients. For an analytic boundary, the Schwarz function produces two natural families of boundary modes. This suggested using a Schwarz pseudo-basis to encode the Dirichlet boundary condition in an infinite Wronskian matrix.

2. **Preliminary numerical investigation.** Before the infinite-dimensional construction was justified, the author implemented finite truncations of the proposed matrix and examined whether its determinant could locate Dirichlet eigenvalues. These experiments helped assess the computational potential of the construction and exposed questions concerning truncation and numerical stability.

3. **Reformulation by harmonic extension.** After the boundedness problem of the operator H was found in the original pseudo-basis formulation, the construction was reorganized on the boundary Hilbert space defined by harmonic measure. A boundary datum is extended harmonically into the domain, and iterated primitive operators generate a Helmholtz solution. Its boundary trace defines the Wronskian operator $W(\lambda)$. The resulting formulation permits the decomposition of $W(\lambda)$ as the identity plus a Hilbert–Schmidt operator and hence the definition of its second regularized determinant.

4. **Representation and kernel correspondence.** It remained necessary to show that the boundary-data construction did not omit relevant Helmholtz solutions. A Vekua-type representation theorem provided the required result and established the correspondence between the kernel of $W(\lambda)$ and the Dirichlet eigenspace.

5. **Construction from conformal data.** The operator was finally represented in conformal Fourier coordinates. This connected $W(\lambda)$ to the coefficients of the normalized Riemann map and produced finite matrices suitable for computation. The construction was examined numerically through forward and local inverse experiments for the Neumann oval.

The author was responsible for the literature study, mathematical derivations, comparison of alternative constructions, numerical implementation, execution of the experiments, production of figures and tables, analysis of the results, and drafting and revision of the report.

(3) Main difficulties and their resolution

1. **Boundedness of the matrix representation.** Passing from the Schwarz pseudo-basis to the conformal Fourier basis introduced a change-of-coordinate matrix V, while the boundedness of V and V^{-1} could not be justified on the required coefficient space. The construction was therefore reformulated on the boundary Hilbert space defined by harmonic measure. In this setting H is the bounded multiplication operator with symbol $|\eta|^2$, and the primitive operators provide the Hilbert–Schmidt estimates required for $W(\lambda) - I$.

2. **Representation of Helmholtz solutions.** The early approach suggested proving completeness of the Schwarz modes. This statement was stronger than required and difficult to establish. The problem was reduced to showing that every relevant Helmholtz solution arises from suitable boundary data. The Vekua-type representation theorem was introduced to complete this step.

3. **Definition of the determinant.** Since $W(\lambda)$ is infinite-dimensional, the ordinary determinant cannot be used directly, which limited its applications in numerical computing. By proving that $W(\lambda) - I$ is Hilbert–Schmidt, the author obtained a well-defined second regularized determinant and a scalar criterion for the Dirichlet spectrum.

4. **Finite-dimensional computation.** The numerical method requires truncating the Riemann mapping coefficients, Fourier modes, operator series, and intermediate matrix products. Several truncation levels were compared, and candidate eigenvalues were located through the smallest singular value and checked against finite-element reference values. A local inverse experiment was also performed to examine the recovery of selected conformal coefficients. These experiments test local computability without establishing global uniqueness or stability.

(4) Disclosure of generative-AI use

Was generative AI used: yes.

The author used OpenAI ChatGPT (mainly 5.6-Sol) as an auxiliary tool for obtaining literature-search leads, discussing possible gaps in proof strategies, checking notation and internal consistency, improving the organization and English expression of drafts, and reviewing the logic of the numerical implementation. It was used intermittently during the theoretical investigation and several times a day during the final organization and revision of the manuscript. Both advisors were informed of and approved this use.

The numerical programs were fully written and executed by the author, and the AI tools did not generate or alter the reported data. Mathematical suggestions were checked through independent derivation, consultation of the cited literature, numerical reproduction, or discussion with the advisors. The author reviewed the final report and accepts responsibility for its mathematical content and presentation. All relevant AI conversation records have been retained for verification.